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FACULTY OF ELECTRICAL ENGINEERING
DEPARTMENT OF CYBERNETICS
MULTI-ROBOT SYSTEMS



Learning an Autonomous Drone Racing Policy via Differentiable Physics

Master's Thesis

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Guidelines:

1. Review and analyze the current state of the art in autonomous drone racing control methods, with emphasis on reinforcement learning and differentiable physics approaches.
2. Implement a differentiable simulation framework for quadrotor UAVs, capturing rigid-body dynamics and basic aerodynamic effects. Leverage differentiability to improve the sample efficiency of training.
3. Design a control architecture in which a learning-based component generates high-level velocity commands and adaptive control gains for a low-level geometric velocity controller.
4. Integrate a geometric velocity controller operating on the Special Orthogonal Group $SO(3)$, capable of handling variable setpoints and gains provided by the higher-level policy.
5. Evaluate the proposed approach in representative simulated environments (e.g., Gazebo) and study the policy's transferability. Optionally, test the learned policy on a real platform.
6. Compare the performance of the proposed method with standard model-free reinforcement learning approaches and classical fixed-gain controllers, assessing metrics such as task completion time, success rate, training efficiency, robustness, and generalization to unseen environments.

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Declaration

I declare that presented work was developed independently, and that I have listed all sources of information used within, in accordance with the Methodical instructions for observing ethical principles in preparation of university theses.

Date 15.08.2026

Adam El Ghaib

Abstract

Todo

Keywords Unmanned Aerial Vehicles, Autonomous Drone Racing, Simulation to Reality Transfer, Reinforcement Learning, Domain Randomization

Abbreviations

ADEs Automatic Differentiation Engines

ADR Autonomous Drone Racing

API Application Programming Interface

BPTT Backpropagation Through Time

CNNs Convolutional Neural Networks

CRNN Convolutional Recurrent Neural Network

CTBR Collective Thrust and Body Rates

DC Direct Current

DFBC Differential Flatness Based Control

DP Dynamic Programming

DQN Deep Q-Learning

DOFs Degrees-Of-Freedom

DR Domain Randomization

DoF Degrees of Freedom

EKF Extended Kalman Filter

ESC Electronic Speed Controller

IMU Inertial Measurement Unit

LiPo Lithium Polymer

ML Machine Learning

MLP Multi-Layer Perceptron

MLPs Multi-Layer Perceptrons

MPC Model Predictive Control

NN Neural Network

OCP Optimal Control Problem

OCPs Optimal Control Problems

POMDP Partially Observable Markov Decision Process

PPO Proximal Policy Optimization

QP Quadratic Programming

RL Reinforcement Learning

RMSProp Root Mean Square Propagation

RMSE Root Mean Square Error

SGD Stochastic Gradient Descent

SR Success Rate

SRT Single Rotor Thrust

Sim2Real Simulation-to-Reality

LVHR Linear Velocities and Heading Rate

TRPO Trust Region Policy Optimization

TWR Thrust to Weight Ratio

UAV Unmanned Aerial Vehicle

VIO Visual Inertial Odometry

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Chapter 1

Introduction

In this chapter we briefly introduce the reader to the sport of Autonomous Drone Racing (ADR), its associated challenges and the current state of the art to adress them. Finally, we provide an overvirew of the structure and objectives of this thesis.

1.1 Background

ADR has emerged as a compelling platform in robotics to develop and deploy novel control techniques in real-world scenarios. The aim of this sport is to navigate through a sequence of gates in minimum time, pushing the platform's capabilities to their limits [1]. To maintain the platform agile and light, the hardware is kept minimal; a typical ADR drone consists of a quadrotor equipped with an RGB camera, an Inertial Measurement Unit (IMU), an onboard computer, a flight controller, brushless Direct Current (DC) motors with propellers, and an Electronic Speed Controller (ESC) all powered by a Lithium Polymer (LiPo) battery [2, 3].

ADR presents several challenges, which can be divided into the following subgroups depicted in Figure 1.1.

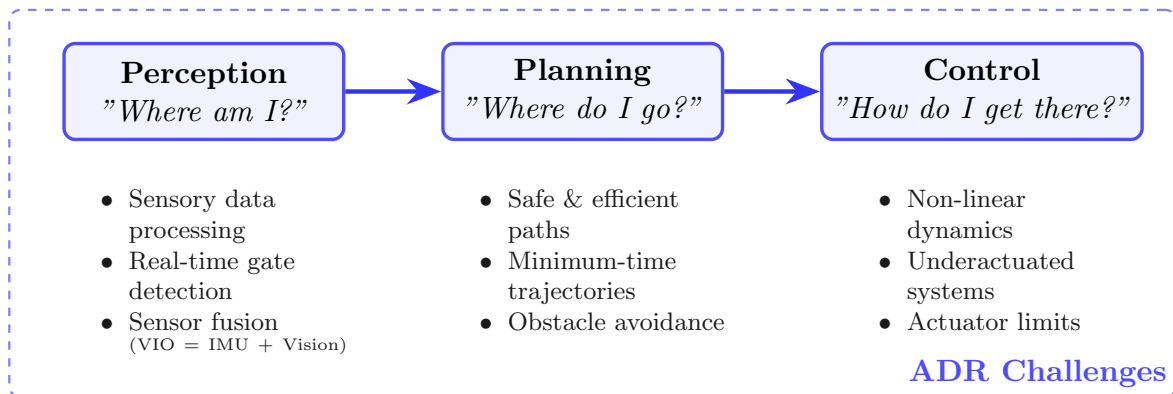


Figure 1.1: Modular navigation stack for robotics and its associated challenges.

The control stack begins at the **perception** level. This stage involves processing sensory data from the camera and IMU, which is inherently noisy and uncertain. The main challenge at this level is to filter and fuse this data to estimate both the position of the gates and the

state of the drone. The fusion of camera and IMU measurements is known as Visual Inertial Odometry (VIO). State estimation is typically performed using filtering techniques that combine a system model with sensor measurements, such as a Kalman Filter [4] or variants [5]. Given an estimate of the drone state and the environment, the next step is **planning**. The objective in ADR is to compute a minimum-time trajectory that passes through the gates while avoiding collisions. Trajectory planning can be computationally demanding, making efficient approaches particularly important. The final stage of the pipeline is **control**. Given a planned trajectory, the goal is to obtain the control inputs to track it with minimal error. This task is challenging due to the highly nonlinear dynamics of the quadrotor. Additionally, the system is underactuated, with only four out of the six Degrees-Of-Freedom (DOFs) directly controllable. Moreover, yaw control authority is typically limited, and actuator saturation further constrains performance.

Many of these challenges are common in robotics, and a variety of methods have been proposed to address them. Learning-based methods, particularly those relying on using a Neural Network (NN), are gaining popularity due to their low-latency inference and robustness to noise. Individual components of the navigation stack can be learned using NNs. For instance, Convolutional Neural Networks (CNNs) can process VIO data to detect gates, while Multi-Layer Perceptrons (MLPs) can be used for planning or control. Combinations of multiple stages, such as perception and planning or planning and control, can also be learned jointly. In some cases, the entire pipeline is learned end-to-end [1]. Learning-based approaches offer several advantages; however, training these models typically requires large amounts of data, and current optimization methods can be sample inefficient. Since collecting such data in real-world settings is often impractical, simulated environments that approximate real-world distributions are commonly used, enabling thousands of parallel training instances. Recent research has explored the use of Automatic Differentiation Engines (ADEs) supported by Machine Learning (ML) frameworks such as [6, 7, 8] to propagate gradients from a cost function to NN parameters. This approach significantly improves sample efficiency [9] and avoids treating the quadrotor dynamics as a black box.

In this thesis, we investigate this direction by focusing on the use of differentiability to learn the control component. We aim to demonstrate that this approach is a promising avenue for research in robotics and particularly for ADR.



Figure 1.2: Long exposure photography in the A2RL 2024 drone racing championship.

1.2 Related Work

The evolution of ADR has progressed from early academic benchmarks in the IROS competitions between 2016 and 2018 [10, 11, 12], where drones reached speeds of only a few meters per second, to highly professionalized racing, where autonomous drones exceeded 150 km/h and outperformed professional human pilots.

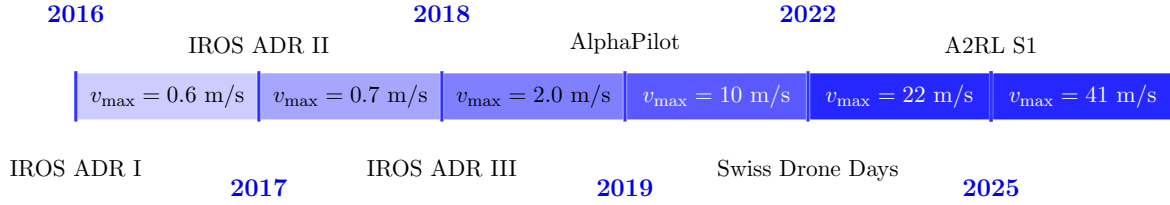


Figure 1.3: Maximum reached speeds over time (adapted from [1]).

Since the AlphaPilot competition [13] in 2019, there has been a clear trend toward incorporating learning-based components into the navigation stack. In [14], the pipeline combined VIO and gate detections within an Extended Kalman Filter (EKF) for state estimation, followed by trajectory planning using sampling-based receding horizon methods. Control was implemented through a cascade of position and attitude controllers, while NNs were used exclusively for visual gate detection. Subsequent work in [15] retained a similar perception pipeline but replaced planning and control with a feedforward Multi-Layer Perceptron (MLP), optimized using Proximal Policy Optimization (PPO) [16].

In a related setup, [17] demonstrated that model-free Reinforcement Learning (RL) could outperform traditional optimal control approaches. The authors attribute the increase in performance of RL due to two main factors. First, the ability of model-free RL to directly formulate the task goal as part of the optimization function without the need of decomposing the the problem into planning and control. This allows their policy to explore a broader range of behaviours that provide robustness to unmodeled dynamics effects. Furthermore, since RL is usually modeled as a Markov Decision Process, the transition model is probabilistic, which allows for Domain Randomization (DR). "By training in randomized environments, the agent becomes more robust and adaptable, making it more effective at handling disturbances and model uncertainty" [17].

On a similar note, recent approaches have explored model-based RL, where a learned world model enables training through imagined rollouts. In [18, 19], the authors adopt an actor-critic architecture based on Dreamer [20]. This approach achieved champion-level performance and has the advantage of directly mapping visual data, raw pixel information, to direct motor commands skipping the need of implementing an intermediate controller. While these methods achieve strong performance, they suffer from high training times and limited sample efficiency; for instance, training in [19] can take up to 50 hours, significantly slowing down the research cycle.

To address these limitations, recent work has shifted toward differentiable approaches that enable backpropagation of gradients from an objective function through the entire navigation stack, this is known as Backpropagation Through Time (BPTT). This paradigm avoids treating the quadrotor dynamics as a black box and substantially improves sample efficiency by leveraging the first-order gradient instead of relying on zeroth-order optimization.

The work of [9] pioneered this research direction, achieving agile vision-based flight in cluttered environments using a Convolutional Recurrent Neural Network (CRNN) and a simplified double-integrator model. Their policy achieves zero-shot generalization to the real-world environment, that is to say, no fine-tuning or retraining is required at policy deployment. In their set up, they run the policy on a budget-friendly microprocessor and a depth camera, allowing for robust obstacle avoidance without needing an IMU or GPS coordinates. Additionally, they deploy the policy simultaneously to different drones and observe emerging behaviours such as, waiting the other agents to cross a gate without explicitly adding collaborative terms on the loss function.

Another differentiable simulation framework is introduced in [21], which has inspired the simulator design used in this work. The simulator supports different training tasks such as hovering, position control, obstacle avoidance and racing scenarios, all differentiable and support parallel environment execution for policy training. Similarly, [22] presents an efficient differentiable quadrotor framework implemented in JAX [7] for learning stabilization from visual features, and compares it against standard PPO. Building on this line of work, [23, 24] demonstrate real-time policy adaptation using real-world data, leveraging the improved sample efficiency of differentiable methods.

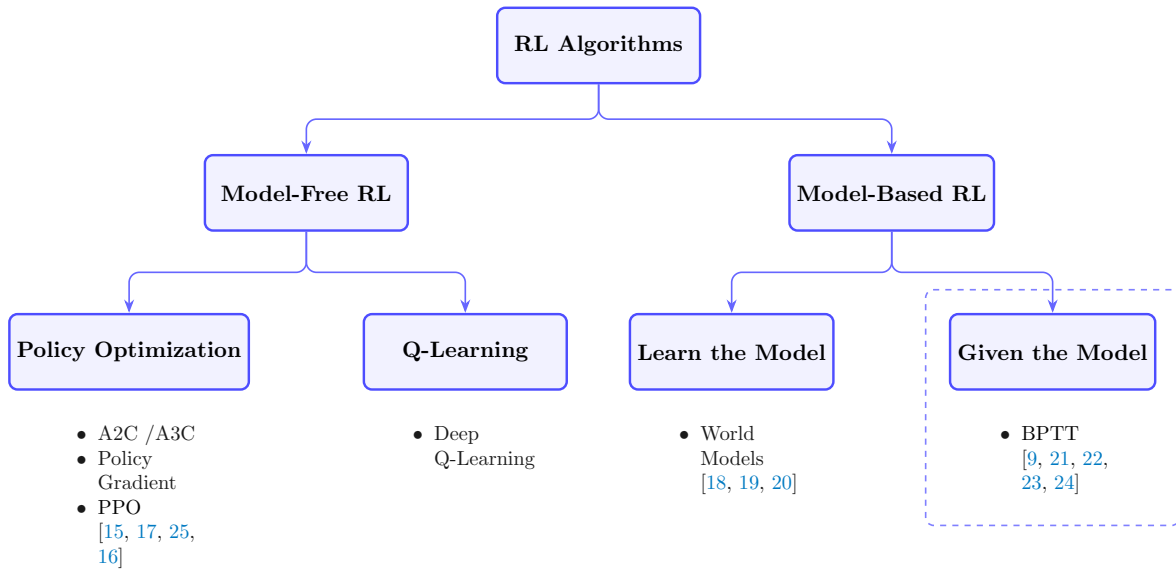


Figure 1.4: Classification of reinforcement learning approaches.

On the other hand, another research direction relevant for this thesis, aims to answer the question of how to tune the gains of a given controller to achieve optimal performance. The authors of [26] show that the faster a trajectory to be tracked is, the more sensitive is the tracking performance to the controller gains. This way, they propose an algorithm based on the Metropolis-Hastings sampling [27], to tune the gains of a Model Predictive Control (MPC) controller for different segments of a trajectory. The trajectory is divided in segments with similar height gradients and minimum time of 2 s per segment. Their approach showed an increase on trajectory tracking success compared to another automatic gain tuning method based on genetic algorithms [28].

1.3 Objectives and Structure

The primary objective of this thesis is to create a differentiable quadrotor simulator to backpropagate the gradients from a given cost function to the NN parameters to command autonomously a drone. A low-level geometric controller will take the velocity and heading rate commands from the network and map them to wrench commands. Additionally, the policy will be augmented to output the gains of the geometric $SO(3)$ controller.

As secondary objectives, we perform an evaluation of the policy network on Gazebo utilizing the MRS UAV stack [29], and we test the controller on a real platform. Furthermore, we perform a comparative study between different control modalities, and, model-free against model-based optimization methods.

Table 1.1: Thesis objectives.

Primary Objectives	
Obj-1.1	Create a differentiable quadrotor simulator.
Obj-1.2	Train a NN that outputs velocity and heading commands.
Obj-1.3	Augment the NN to output adaptive gains for the $SO(3)$ controller.
Secondary Objectives	
Obj-2.1	Evaluate how other control modalities compare to geometric control.
Obj-2.2	Compare model-based optimization against model-free.
Obj-2.3	Evaluate the fixed-gain controller against the adaptive gain controller.
Obj-2.4	Assess the transferability of the policies to other environments.

The thesis is divided in 6 chapters. In Chapter 1 provide an introduction of the thesis topic and cover the most relevant related work. In Chapter 2, we perform In Chapter 3, we present the methodology of our method, we explain the learning environment, the quadrotor model, the control modalities that have been tested, the domain randomization strategy and we formulate the cost function to be minimized. In Chapter 4, we show the results of our method. We analyze the effect of the different control abstractions on the sample efficiency and performance. Additionally we show the effect of other implementation decisions and we analyze real world flight data. In Chapter 5, we discuss the results of our method. Finally, in Chapter 6 we state our future research directions.

Chapter 2

Preliminaries

This chapter covers relevant theory on which this thesis is built. First, in Section 2.1 we explain what is an Optimal Control Problem (OCP) and we formulate the OCP for ADR. Then, in Section 2.2 and Section 2.3, we cover some classical and learning-based approaches to adress the OCP. Finally, in Section 2.4 we explain how automatic differentiation can be used to optimize a control policy.

2.1 Optimal Control Problem

According to [30], optimal control was born in 1687 in the Netherlands when Johan Bernoulli published his solution to the brachystochrone problem. Optimal control can be regarded as an extension of calculus of variations, with the aim of finidng a control policy that minimizes a cost function. In this section, we formulate optimal control problems related to this thesis.

2.1.1 General Discrete-Time Optimal Control Problem

In general, a nonlinear discrete-time OCP can be formulated as the optimization of a running cost (or loss) function $L_k()$, plus a terminal cost function $\phi()$, subject to the system's dynamics $\mathbf{f}_k()$, and the set of permitted states $\mathcal{X}_k$, and controls $\mathcal{U}_k$.

$$\begin{aligned} \underset{\mathbf{X}, \mathbf{U}}{\text{minimize}} \quad & \left(\phi(\mathbf{x}_N, N) + \sum_{k=0}^{N-1} L_k(\mathbf{x}_k, \mathbf{u}_k) \right) \\ \text{subject to} \quad & \mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, N-1, \\ & \mathbf{u}_k \in \mathcal{U}_k, \quad k = 0, \dots, N-1, \\ & \mathbf{x}_k \in \mathcal{X}_k, \quad k = 0, \dots, N, \end{aligned} \tag{2.1}$$

Where

- N is the final discrete time,
- $\mathbf{x}_k \in \mathbb{R}^n$ is the state at the discrete time $k \in \mathbb{Z}$,
- $\mathbf{u}_k \in \mathbb{R}^m$ is the control at the discrete time $k \in \mathbb{Z}$,
- $\mathbf{X} = [\mathbf{x}_0, \dots, \mathbf{x}_N]$ is the trajectory of states,
- $\mathbf{U} = [\mathbf{u}_0, \dots, \mathbf{u}_{N-1}]$ is the trajectory of control actions.

2.1.2 Minimum-Time Optimal Control Problem

The last formulation of an OCP can be further specified for the case of drone racing in several manners. Since the main objective is to complete a lap through the racing track as fast as possible, this problem can be written as minimum-time problem.

$$\begin{aligned}
& \underset{\mathbf{U}, N}{\text{minimize}} && \sum_{k=0}^{N-1} 1 \\
& \text{subject to} && \mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, N-1, \\
& && \mathbf{u}_k \in \mathcal{U}_k, \quad k = 0, \dots, N-1, \\
& && \mathbf{x}_k \in \mathcal{X}_k, \quad k = 0, \dots, N,
\end{aligned} \tag{2.2}$$

The racing track restrictions, i.e. passing through a sequence of gates while avoiding obstacles, can be introduced as hard constraints under the set of permitted states $\mathcal{X}_k$ as

$$\mathcal{X}_k = \{\mathbf{x}_k \in \mathbb{R}^n \mid \mathbf{p}_k \in \mathcal{G}_{g(k)}, \quad \mathbf{p}_k \notin \mathcal{O}_j, \quad \forall j \in \{1, \dots, M\}\}, \tag{2.3}$$

where,

- $\mathbf{p}_k \in \mathbb{R}^3$ is the position of the drone at time step k ,
- $\mathcal{G}_{g(k)}$ is the feasible region defined by gate g at time step k , with $g(k) \in \{1, \dots, G\}$ denoting the index of the gate to be passed at k ,
- G is the total number of gates in the racing track,
- $\mathcal{O}_j$ is the region occupied by the j -th obstacle,
- M is the total number of obstacles.

2.1.3 Simplified Minimum-Time Optimal Control Problem

The original problem formulation (Section 2.1.2) is inherently complex, as it requires simultaneously solving for an optimal trajectory that clears all gates and avoids obstacles while determining the optimal control sequence achievable within the vehicle's dynamics and actuator constraints. To make the problem more tractable, this thesis assumes that a reference trajectory or path is provided a priori. Consequently, the OCPs addressed hereafter focus on trajectory tracking, formulated as follows:

Trajectory Tracking

In the trajectory tracking OCP, the objective is to find the optimal control sequence $\mathbf{U}$ that minimizes the error between a given reference state $\mathbf{x}_k^{\text{ref}}$ and the current system state $\mathbf{x}_k$, subject to dynamics, actuator, and state constraints. While this simplifies the optimization significantly it comes at the cost of reduced robustness. By assuming the trajectory is fixed, the system cannot re-optimize the path to accommodate state constraints or environmental changes, effectively limiting the drone's agility compared to a unified spatial-temporal optimization.

$$\begin{aligned}
& \underset{\mathbf{U}}{\text{minimize}} && \sum_{k=0}^{N-1} \left\| \mathbf{x}_k - \mathbf{x}_k^{\text{ref}} \right\|_{\mathbf{Q}}^2 \\
& \text{subject to} && \mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, N-1, \\
& && \mathbf{u}_k \in \mathcal{U}_k, \quad k = 0, \dots, N-1, \\
& && \mathbf{x}_k \in \mathcal{X}_k, \quad k = 0, \dots, N,
\end{aligned} \tag{2.4}$$

Gate Progress

An alternative objective to reduce the complexity without sacrificing much on robustness is to optimize the progress toward the target gate position. Unlike trajectory tracking, which imposes a strict temporal constraint for every spatial coordinate, the gate progress OCP leaves the path and speed toward the gate as a decision variable. This allows the controller to dynamically adjust to the current vehicle's state providing more flexibility.

$$\begin{aligned}
& \underset{\mathbf{U}}{\text{minimize}} && \sum_{k=1}^N \left(\left\| \mathbf{p}_k^{\mathcal{W}} - \mathbf{p}_{g_k}^{\mathcal{W}} \right\|_2 - \left\| \mathbf{p}_{k-1}^{\mathcal{W}} - \mathbf{p}_{g_k}^{\mathcal{W}} \right\|_2 \right) \\
& \text{subject to} && \mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, N-1, \\
& && \mathbf{u}_k \in \mathcal{U}_k, \quad k = 0, \dots, N-1, \\
& && \mathbf{x}_k \in \mathcal{X}_k, \quad k = 0, \dots, N,
\end{aligned} \tag{2.5}$$

2.2 Classical Control approaches

Given the OCP formulated in last sections, by solving the quadratic program, a trajectory of control inputs that minimizes the cost function can be obtained. However, in this direct approach, the control is open-loop. Hence it is not robust to external disturbances and model mismatches. In the classical literature, a well established approach that addresses this issue is an MPC. Another common control approach is Differential Flatness Based Control (DFBC), "The states and the inputs can be written as algebraic functions of four carefully selected flat outputs and their derivatives" [31]. Leveraging this property, motor command can be directly obtained from a given trajectory to be tracked.

Many other quadrotor control methods are well established on the literature, ranging from Proportional Derivative Integral (PID) controllers, Linear Quadratic Regulators (LQR), $\mathcal{L}_2$ adaptive control, H_∞ norm minimization, μ -synthesis and many others, each with their own advantages and disadvantages. However, in this section we give an overview of MPC and DFBC as we find them the most relevant for this thesis.

2.2.1 Model Predictive Control

Receding horizon control or also known as MPC is a direct approach to solving an OCP. The key idea behind this method is to solve the control problem for a given prediction horizon of N_p steps. As a solution we obtain the control inputs that minimize the cost function for that horizon, nonetheless we only take $N_c \sim 2$ or 3 steps, of those controls and discard the rest. Then, we solve again the control problem with the updated states, and repeat the process.

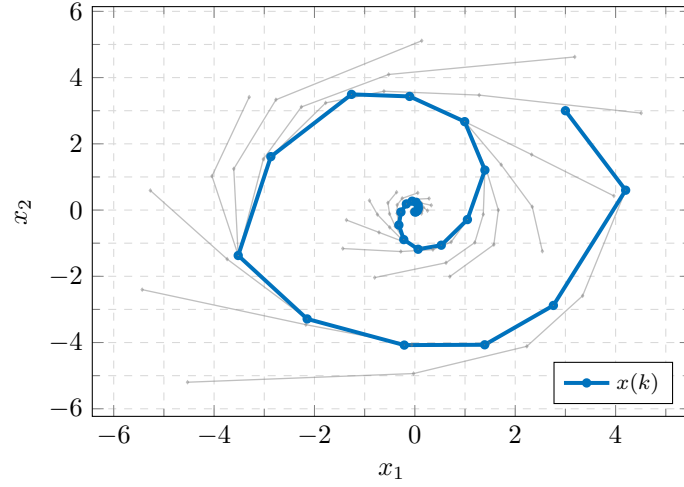


Figure 2.1: Example of regulation using model predictive control (example from [32]).

The main advantage of this method is that it remains robust to disturbances and handles well constraints on the control inputs and states. However, it is computationally costly and can present numerical convergence issues.

2.2.2 Differential Flatness Based Control

As shown in [31], the quadrotor's state and control inputs can be expressed as algebraic function of carefully selected flat outputs. Choosing the position of the center of mass $\mathbf{r} = [r_x, r_y, r_z]^\top$ and its heading angle η as outputs, the quadrotor's $\hat{\mathbf{b}}_3$ unit vector has the same direction as its thrust vector which corresponds to its acceleration,

$$\mathbf{t} = \ddot{\mathbf{r}} + [0, 0, g]^\top \quad (2.6)$$

$$\hat{\mathbf{b}}_3 = \frac{\mathbf{t}}{\|\mathbf{t}\|}. \quad (2.7)$$

From the heading angle, the heading vector $\mathbf{h}$ can be obtained, and subsequently the $\hat{\mathbf{b}}_2$ and $\hat{\mathbf{b}}_1$ axis,

$$\mathbf{h} = [\cos \eta, \sin \eta, 0]^\top \quad (2.8)$$

$$\hat{\mathbf{b}}_2 = \frac{\hat{\mathbf{b}}_3 \times \mathbf{h}}{\|\hat{\mathbf{b}}_3 \times \mathbf{h}\|} \quad (2.9)$$

$$\hat{\mathbf{b}}_1 = \hat{\mathbf{b}}_2 \times \hat{\mathbf{b}}_3 \quad (2.10)$$

which defines the attitude of the drone, $\mathbf{R}(\phi, \theta, \psi) = [\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3]$. From Newton's equation of motion we have that the translational dynamics are,

$$m\ddot{\mathbf{r}} = -mg\hat{\mathbf{e}}_3 + T_c\hat{\mathbf{b}}_3 \quad (2.11)$$

and from Euler equation, we get the rotational dynamics,

$$\boldsymbol{\tau} = \mathbf{J}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} \quad (2.12)$$

where $\mathbf{J}$ is the inertia matrix and $\dot{\boldsymbol{\omega}}$ are the body angular rates. The first derivative of Equation 2.11 gives,

$$m \ddot{\mathbf{r}} = \dot{T}_c \hat{\mathbf{b}}_3 + \boldsymbol{\omega} \times u_1 \hat{\mathbf{b}}_3. \quad (2.13)$$

The change in collective thrust $\dot{T}_c$ corresponds to the projection $\dot{T}_c = \hat{\mathbf{b}}_3 \cdot m \ddot{\mathbf{r}}$, substituting in Equation 2.13, we can obtain the vector $\mathbf{h}_\omega = \boldsymbol{\omega} \times \hat{\mathbf{b}}_3$,

$$\mathbf{h}_\omega = \frac{m}{u_1} (\ddot{\mathbf{r}} - (\hat{\mathbf{b}}_3 \cdot \ddot{\mathbf{r}}) \hat{\mathbf{b}}_3) \quad (2.14)$$

which allows to obtain the angular velocities. To completely define the quadrotor's state we need the angular accelerations $\dot{\boldsymbol{\omega}}$ which can be obtained from the second derivative of Equation 2.11. Hence, the control inputs are function of the flat outputs as $u_1 = T_c = m\|\mathbf{t}\|$, and $[u_2, u_3, u_4]^\top = \boldsymbol{\tau}$.

This property is exploited in Section 3.2.3 to determine the attitude of the quadrotor and design a geometric controller.

2.3 Learning-Based approaches

Learning-based approaches can be understood as optimization methods in which the optimization process is done by interacting with the environment [33]. There are two main types of RL, model-based and model-free, each have with their own advantages and drawbacks. Next, we give an overview model-free and model-based RL and the most relevant algorithms in the literature of each.

2.3.1 Model-Free Reinforcement Learning

In model-free RL, the transition dynamics of the environment are not explicitly modeled or known a priori. Instead, the policy is optimized directly by gathering experience through interaction with the environment, typically utilizing first-order policy gradient methods. Next, we review three model-free RL approaches; Q-Learning, and two Policy Optimization methods; Policy Gradient and PPO.

Q-Learning

In Q-learning, the objective is to estimate the value of taking a given action at a particular state and store these values in a Q-table. During training, Bellman's equation is used to iteratively update the entries of the table. Once training converges, an optimal policy π^* can be obtained, since the Q-table provides the action with the highest expected return for each state [34].

$$\pi^*(s) = \arg \max_a Q^*(s, a) \quad (2.15)$$

Q-learning can be interpreted as a formulation of Dynamic Programming (DP). For simple problems, such as finding the optimal strategy to escape a maze, this approach performs well. However, for high-dimensional or complex environments, storing and updating the Q-table becomes computationally infeasible.

To overcome this limitation, Deep Q-Learning (DQN) was introduced in [35]. Instead of explicitly storing the Q-values in a table, a neural network is trained to approximate the action-value function.

Policy Gradient

In policy optimization methods, instead of learning the action-value function and deriving an optimal policy by selecting the best action at a given state, the policy is learned directly through the parameters of a neural network θ [34]. These parameters are optimized with respect to an objective function $J(\theta)$ that encodes the desired behavior of the policy.

$$J(\theta^*) = \arg \max_{\theta} \mathbb{E}_{\tau \sim \pi_{\theta}} [R(\tau)] \quad (2.16)$$

In this framework, policy optimization methods aim to find the optimal parameters θ^* such that the resulting policy $\pi_{\theta}(a|s)$ selects actions $a \in \mathcal{A}$ for states $s \in \mathcal{S}$ that maximize the expected return over trajectories $\tau = \{s_0, a_0, s_1, a_1, \dots\}$.

Proximal Policy Optimization

This policy optimization method was introduced in 2017 in [16] and has become a standard algorithm in RL. The algorithm addresses instability issues present in policy gradient methods. Differentiating Equation 2.16 gives,

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{s, a \sim \pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi}(s, a)]. \quad (2.17)$$

In practice, Q^{π} is replaced by an advantage estimate A^{π} to reduce variance. Using this vanilla gradient can lead to excessive parameter updates that destabilize the policy. Previously, this issue was addressed in [36] where the authors introduce the Trust Region Policy Optimization (TRPO) which fixes a hard KL-divergence constrain. Note that KL-divergence measures the difference between two data distributions $D_{KL}(P||Q) = \mathbb{E}_x \log \frac{P(x)}{Q(x)}$.

$$\begin{aligned} & \underset{\theta}{\text{maximize}} \quad \mathbb{E} \left[\frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} A_t \right] \\ & \text{subject to} \quad D_{KL}(\pi_{\theta_{old}}||\pi_{\theta}) \leq \delta. \end{aligned} \quad (2.18)$$

However, this method involves the calculation of the Hessian, a second-order derivative matrix which is computationally costly. Instead of imposing a hard constrain the authors of [16] propose two relaxation methods.

(i) Adaptive KL Penalty Coefficient

The cost function becomes,

$$L^{KL PEN}(\theta) = \hat{\mathbb{E}}_t \left[\frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \hat{A}_t - \beta \text{KL}[\pi_{\theta_{old}}(\cdot|s_t), \pi_{\theta}(\cdot|s_t)] \right] \quad (2.19)$$

where β controls the weight of the penalization when the new update deviates too much from the old policy parameters.

(ii) **Clipped Surrogate Objective**

Instead of penalizing the KL divergence, the probability ratio $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is clipped in the cost function.

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2.20)$$

where epsilon is a hyperparameter.

The authors claim that the clipped surrogate objective approach performs better. Since PPO is usually implemented as an actor-critic algorithm a loss term for the critic, $L_t^{VF}(\theta)$, is incorporated, and to encourage exploration, an entropy bonus, $S[\pi_\theta](s_t)$, is added.

Hence, the final PPO loss is,

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t \left[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t) \right] \quad (2.21)$$

where c_1 and c_2 are weighting factors.

2.3.2 Model-Based Reinforcement Learning

Model-based RL uses information of a model to optimize the policy parameters. The model can be predefined, or it can be learned by a neural network.

Given The Model

When the model is known, that is to say, the state equation $\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k)$. The RL architecture can be straightforward, the gradient from the cost function J to the policy parameters can be computed directly by applying the chain rule. And the policy is optimized backpropagating through time the accumulated gradient. In section 2.4, we explain in more detail how the optimization process works as this is the method we use in our methodology to learn the drone racing policies.

Having knowledge of the model increases the sample efficiency. However, if the model is not suitable error can accumulate leading to suboptimal performance. A perfect model is not necessary, but it has to capture the important behaviours of the system. Additionally, NNs tend to overfit to the training data distribution. To prevent overfitting, it is recommended to randomize model parameters, as we explain in Section 3.4.

Learn The Model

An alternative approach involves learning the state transition dynamics directly from environmental interactions to estimate the state equation $\hat{\mathbf{x}}_{k+1} = \hat{f}_\theta(\mathbf{x}_k, \mathbf{u}_k)$. A notable framework in this domain is the Dreamer architecture [20]. Rather than modeling the world in the raw observation space, Dreamer utilizes an encoder to map observations $\mathbf{o}_k$ into stochastic latent representations $\mathbf{z}_k$. The system's dynamics are then captured by a recurrent state-space model: $\mathbf{h}_{k+1} = f(\mathbf{h}_k, \mathbf{z}_k, \mathbf{u}_k)$. This approach offers two primary advantages: the recurrent state provides essential temporal memory, while the mapping to a stochastic latent space facilitates the processing of high-dimensional inputs, such as raw images.

Once the world model is learned, an actor-critic agent can be trained entirely within this learned environment. This paradigm shares conceptual similarities with Model Predictive

Control (MPC), as it allows the agent to account for the impact of current actions on future states. In [19], the authors employ a similar architecture to train an Autonomous Drone Racing (ADR) policy that maps pixels directly to motor commands. Their method demonstrated champion-level performance; however, a drawback of this method is its sample inefficiency, as the authors report that training can require up to 50 hours.

2.4 Differentiable Optimization

Under the paradigm of differentiable optimization, we no longer treat the quadrotor model as a "black-box". The main idea of this thesis is to leverage an automatic differentiation engine to compute the exact analytical gradient of the cumulative loss J_k with respect to the control policy network. In this section, we provide an overview on how the optimization process works.

2.4.1 Computational Graph and Backpropagation

The gradient of the cumulative stage cost function J_k with respect to the network parameters θ is given by the chain rule as,

$$\nabla_{\theta} J_k = \frac{1}{B} \frac{1}{H} \sum_{i=1}^B \sum_{h=k}^{k+H-1} \frac{\partial \mathcal{L}_h}{\partial \mathbf{x}_h} \frac{\partial \mathbf{x}_h}{\partial \mathbf{u}_{h-1}} \frac{\partial \mathbf{u}_{h-1}}{\partial \theta} \quad (2.22)$$

where $\frac{\partial \mathcal{L}_h}{\partial \mathbf{x}_h}$ is the gradient of the stage loss with respect to the state, $\frac{\partial \mathbf{x}_h}{\partial \mathbf{u}_{h-1}}$ encodes the change in the dynamics regarding the applied controls, and $\frac{\partial \mathbf{u}_{h-1}}{\partial \theta}$ is the gradient of the control output with respect to the policy parameters. Since the gradient is accumulated through time due to the recursive dependence of the state on previous states and control inputs, this procedure is known as BPTT.

To compute this gradient analytically, an automatic differentiation engine is used. Many ML frameworks, such as PyTorch [6], TensorFlow [8], and JAX [7], among others, provide autodiff capabilities. At present, JAX is regarded as a high-performance framework for numerical ML computing. However, for this simulator, PyTorch has been selected due to its intuitive interface, extensive documentation, and lower implementation complexity.

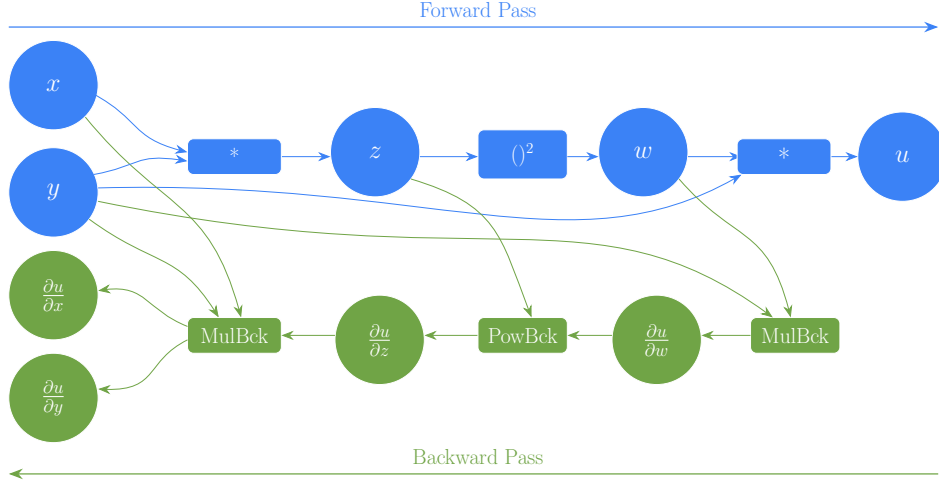


Figure 2.2: Computational graph of the numerical operations of the forward and backward passes.

To illustrate how gradients are computed, consider the next function $f(x, y) = y(xy)^2$. Its gradient is $\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2xy^3 & 3x^2y^2 \end{bmatrix}$ and evaluating at $x = 1$ and $y = 2$, we get $\nabla f(1, 2) = \begin{bmatrix} 16 & 12 \end{bmatrix}$. In PyTorch, this gradient is computed using a computational graph which is constructed during the forward pass. Each mathematical operation is stored as a node in this graph. During backpropagation, PyTorch applies automatic differentiation to traverse the graph from the output back to the input and applying the chain rule to accumulate gradients. The computational graph shown in Figure 2.2 represents this process and Algorithm 1 presents the pseudocode. In practice, the calculation of the gradient would be done simply by calling `u.backward()` in PyTorch.

Algorithm 1 Example Backpropagation for the function $u(x, y) = y(xy)^2$

Require: $x \in \mathbb{R}, y \in \mathbb{R}$

Ensure: $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$

— **Forward Pass (Build Computational Graph)** —

$z \leftarrow x \times y$

▷ Node: MulBackward

$w \leftarrow z^2$

▷ Node: PowBackward

$u \leftarrow w \times y$

▷ Output Node: MulBackward

— **Backward Pass (Reverse-Mode Autodiff)** —

Initialize $\frac{\partial u}{\partial u} \leftarrow 1$

$\frac{\partial u}{\partial w} \leftarrow \frac{\partial u}{\partial u} \cdot y = 1 \cdot 2 = 2$

▷ Backprop through $u = w \cdot y$

$\frac{\partial u}{\partial y} += \frac{\partial u}{\partial u} \cdot w = 1 \cdot 4 = 4$

▷ Accumulate y gradient from this node

$\frac{\partial u}{\partial z} \leftarrow \frac{\partial u}{\partial w} \cdot 2z = 2 \cdot 2 = 4$

▷ Backprop through $w = z^2$

$\frac{\partial u}{\partial x} \leftarrow \frac{\partial u}{\partial z} \cdot y = 4 \cdot 2 = 8$

▷ Backprop through $z = x \cdot y$

$\frac{\partial u}{\partial y} += \frac{\partial u}{\partial z} \cdot x = 4 \cdot 1 = 4$

▷ Accumulate y gradient from this node

— **Results** —

$\frac{\partial u}{\partial x} = 16, \quad \frac{\partial u}{\partial y} = 4 + 4 + 4 = 12$

Additionally, PyTorch allows to customize the computational graph in a manner that the forward and backward pass do not need to be identical. This is known as gradient shaping

or surrogate gradient, and we use this feature in our methodology to skip the motor dynamics.

2.4.2 Optimization Process

Given the gradient $\nabla_{\theta}J$ computed via BPTT the policy parameters are updated using the Adaptive Moment Estimation (Adam) optimizer [37]. Adam is a variant of Stochastic Gradient Descent (SGD). In standard SGD the model parameters are updated proportionally to the gradient in its opposite direction,

$$\theta_{k+1} = \theta_k - \alpha \nabla_{\theta} J_k \quad (2.23)$$

where α indicates the learning rate. The main disadvantages of standard SGD is that near the optimum, the gradient decreases requiring more steps to converge and, that each parameter is updated equally. Adam solves this issues by combining two other variations of SGD: Momentum and Root Mean Square Propagation (RMSProp).

- Momentum accumulates an average of past gradients sustaining progress along the directions of consistent descent and dampening oscillations in high curvature directions.
- RMSProp averages the square of the gradients, which are used to scale the learning rate of each parameter individually, hence, parameters with high historical gradients receive smaller updates than those with smaller gradients

Adam unifies these two concepts through the first and second moment estimates, m and v_k respectively, which are updated as,

$$m_k = \beta_1 m_{k-1} + (1 - \beta_1) \nabla_{\theta} J_{k-1} \quad (2.24)$$

$$v_k = \beta_2 v_{k-1} + (1 - \beta_2) (\nabla_{\theta} J_{k-1})^2 \quad (2.25)$$

where β_1 and β_2 are hyperparameters usually set to 0.9 and 0.999 respectively. Since both moments are initialized at zero $m_0 = v_0 = 0$, the estimates are biased towards zero in the early iterations of training. For this reason, each moment is normalized to,

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^k} \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^k}. \quad (2.26)$$

The final parameter update is given by,

$$\theta_{k+1} = \theta_k - \alpha \frac{\hat{m}_k}{\sqrt{\hat{v}_k} + \epsilon} \quad (2.27)$$

where $\epsilon \ll 1$ is a small constant introduced for numerical stability.

Chapter 3

Methodology

This chapter presents the methodology used in this thesis to train the neural control policies. In Section 3.1, we define the quadrotor model and in Section 3.2 presents the four different control abstractions that have been implemented. Next, in Section 3.4 we explain how the Simulation-to-Reality (Sim2Real) gap is minimized by using DR. Section 3.3, outlines the training objective used to adress the ADR challenge. Lastly in Section 3.5, we summarize the learning framework and we present the final training algorithm.

3.1 Quadrotor Model

In the simulated environments, the quadcopter is modeeld as a 6 Degrees of Freedom (DoF) rigid body. The inertial world frame $\mathcal{W}$ is defined by the fixed orthonormal basis $\{\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_3\}$ while the non-inertial body frame $\mathcal{B}$ by the orthonormal basis $\{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2, \hat{\mathbf{b}}_3\}$, which is attached to the vehicle's center of mass. The state of the Unmanned Aerial Vehicle (UAV) is given by $\mathbf{x} = [\mathbf{p}, \mathbf{v}, \mathbf{q}, \boldsymbol{\omega}, \boldsymbol{\Omega}]^\top$ where the position $\mathbf{p} \in \mathbb{R}^3$ and velocity $\mathbf{v} \in \mathbb{R}^3$ are represented in the world frame $\mathcal{W} \in \mathbb{R}^3$. Conversely, the body rates $\boldsymbol{\omega} \in \mathbb{R}^3$ are represented in the body frame $\mathcal{B} \in \mathbb{R}^3$ and $\boldsymbol{\Omega} \in \mathbb{R}^4$ is a vector of scalar values indicating the rotor speeds. The relationship between the two frames is given by the rotation mapping $\mathbf{b}_i = \mathbf{R}\mathbf{e}_i$, where $\mathbf{R} \in SO(3)$.

The kinematics and dynamics of the quadcopter can be described by the following set of equations:

$$\dot{\mathbf{p}}^{\mathcal{W}} = \mathbf{v}^{\mathcal{W}} \quad (3.1)$$

$$\dot{\mathbf{q}}^{\mathcal{W}} = \frac{1}{2} \mathbf{q}^{\mathcal{W}} \odot \begin{bmatrix} 0 \\ \boldsymbol{\omega}^{\mathcal{B}} \end{bmatrix} \quad (3.2)$$

$$\dot{\mathbf{v}}^{\mathcal{W}} = \frac{\mathbf{R}(\mathbf{q})}{m} (\mathbf{F}_T^{\mathcal{B}} + \mathbf{F}_D^{\mathcal{B}}) + \mathbf{g}^{\mathcal{W}} \quad (3.3)$$

$$\dot{\boldsymbol{\omega}}^{\mathcal{B}} = \mathbf{J}^{-1} (\boldsymbol{\tau}^{\mathcal{B}} - \boldsymbol{\omega}^{\mathcal{B}} \times \mathbf{J}\boldsymbol{\omega}^{\mathcal{B}}) \quad (3.4)$$

$$\dot{\boldsymbol{\Omega}} = \frac{1}{\tau_m} (\boldsymbol{\Omega}_{\text{cmd}} - \boldsymbol{\Omega}) \quad (3.5)$$

In these expressions, $\odot$ denotes the quaternion multiplication operator, $\mathbf{R}(\mathbf{q})$ represents the rotation matrix mapping the body frame $\mathcal{B}$ to the world frame $\mathcal{W}$. The vectors $\mathbf{F}_T^{\mathcal{B}}$ and $\mathbf{F}_D^{\mathcal{B}}$ correspond to the collective thrust and drag forces, respectively. Finally, $\mathbf{J}$ is the diagonal inertia matrix, τ_m is the motor time constant, and $\mathbf{g}^{\mathcal{W}}$ is the gravity vector.

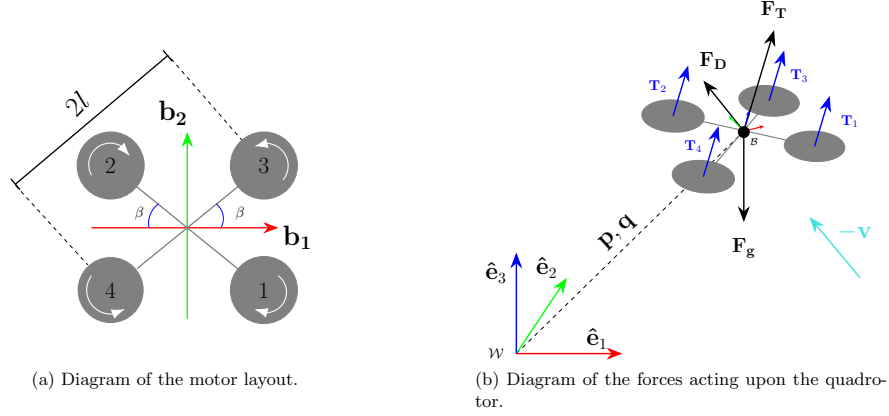


Figure 3.1: Quadrotor configuration and force diagram.

In the body frame $\mathcal{B}$, the individual motor thrusts T_i , $i = 1, \dots, 4$, are assumed to be perpendicular to the $\{\hat{\mathbf{b}}_1, \hat{\mathbf{b}}_2\}$ plane. The magnitudes of collective thrust F_T , and body torques $\tau^{\mathcal{B}}$, can be obtained from Equation 3.6. To be strictly rigorous, the torques produced by the rotor's angular acceleration and gyroscopic effects should be accounted for. Where $\mathbf{\Gamma}$ is the allocation matrix. Nonetheless, these effects produced by the rotor speeds, can be neglected [38] and the mapping simplifies to,

$$\begin{bmatrix} F_T \\ \tau_{b1} \\ \tau_{b2} \\ \tau_{b3} \end{bmatrix} = \overbrace{\begin{bmatrix} 1 & 1 & 1 & 1 \\ -l \sin \beta & l \sin \beta & l \sin \beta & -l \sin \beta \\ -l \cos \beta & l \cos \beta & -l \cos \beta & l \cos \beta \\ -c_{tf} & -c_{tf} & c_{tf} & c_{tf} \end{bmatrix}}^{\mathbf{\Gamma}} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} \quad (3.6)$$

where l is the arm length, β is the arm angle, and c_{tf} the linear torque constant. The individual motor thrusts T_i , can be obtained from the rotor speeds Ω_i with the thrust map coefficients, $\mathbf{k}_T = [a_1, a_2, a_3]$ which can be obtained experimentally by fitting a quadratic curve to the rotor speed versus thrust data.

$$T_i = \mathbf{k}_T \begin{bmatrix} \Omega_i^2 \\ \Omega_i \\ 1 \end{bmatrix} = [a_0 \ a_1 \ a_2] \begin{bmatrix} \Omega_i^2 \\ \Omega_i \\ 1 \end{bmatrix}, \quad i = 1, \dots, 4 \quad (3.7)$$

The terms a_1 and a_2 can be neglected, hence Equation 3.7 is simplified to $T_i = a_0 \Omega_i^2$.

To complete the quadrotor model, the aerodynamic drag force $\mathbf{F}_D^{\mathcal{B}}$ is incorporated using a simplified quadratic formulation. Other aerodynamic effects, such as rotational drag and ground interaction, are neglected. The drag force is expressed as:

$$\mathbf{F}_D^{\mathcal{B}} = -\frac{1}{2} \rho \|\mathbf{v}^{\mathcal{B}}\| \mathbf{C}_D \mathbf{A} \mathbf{v}^{\mathcal{B}} \quad (3.8)$$

where ρ denotes the air density, $\mathbf{A} = \text{diag}(A_{b1}, A_{b2}, A_{b3})$ represents the reference area along each body axis, and $\mathbf{C}_D = \text{diag}(C_{D_{b1}}, C_{D_{b2}}, C_{D_{b3}})$ is a diagonal matrix of non-dimensional drag coefficients. The velocity must be expressed in the body frame, obtained as $\mathbf{v}^{\mathcal{B}} = \mathbf{R}(\mathbf{q})^\top \mathbf{v}^{\mathcal{W}}$.

3.2 Control Abstraction

In this thesis, we refer to the control abstraction or control mode as the modality of the policy outputs used to control the quadrotor. The choice of control abstraction affects both the performance of the controller for a given task and the sample efficiency during training. For this reason, it is worthwhile to investigate and implement multiple control abstractions.

In this thesis, three control modalities are considered. For all control modalities, the policy output layer pass through a sigmoid activation function, Equation 3.9, ensuring that the action space is bounded, which facilitates stable learning.

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (3.9)$$

The lowest level of abstraction consists of directly outputting Single Rotor Thrust (SRT) for each motor. Then, we implement an intermediate level of abstraction that outputs collective thrust and the desired body rates, which are subsequently mapped to individual rotor thrusts through the control allocation matrix Equation 3.6. The highest level of abstraction outputs desired linear velocities and heading rate, which are converted into wrench using an SO(3) geometric controller, inspired by [39], and subsequently mapped to rotor thrusts in a manner analogous to the collective thrust and body rate formulation. The mathematical formulation of these transformations is presented next.

3.2.1 Single Rotor Thrust

When the SRT control mode is chosen, the only transformation to be applied is scaling the policy output $\mathbf{u} \in [0, 1] \in \mathbb{R}^4$ to the actuator limits, $\mathbf{T}_{\text{cmd}} \in [T_{\min}, T_{\max}]$. Therefore, the transformation is as simple as:

$$\mathbf{T}_{\text{cmd}} = \mathbf{u} (T_{\max} - T_{\min}) + T_{\min} \quad (3.10)$$

Once the scaling is applied, to better emulate the behaviour of a real motor, a first order model with rotor speed saturation is implemented. First, the commanded motor thrusts $\mathbf{T}_{\text{cmd}}$ are converted to rotor speeds with,

$$\mathbf{T}_{\text{cmd}} \approx k \boldsymbol{\Omega}_{\text{cmd}}^2. \quad (3.11)$$

Then the commanded rotor speeds $\boldsymbol{\Omega}_{\text{cmd}}$, are passed through the following first order transfer function Equation 3.12, and saturated according to the maximum rotor speed rate, $\dot{\Omega}_{\text{cmd}}$ given by the manufacturer.

$$H(s) = \frac{1}{\tau_m s + 1} \quad (3.12)$$

Where τ_m is the motor time constant. Finally, the rotor speeds obtained after the described process are mapped again to single rotor thrust values using Equation 3.11 and then converted to collective force and torques with Equation 3.6, to calculate the next.

Modeling the motor dynamics provides to the policy with more realistic. However, for training purposes it has been decided to skip the motor dynamics during the backwards pass

as including them, significantly destabilizes gradients and difficults the optimization process. Particularly, the mapping of Equation 3.11 produces high variations on the gradient, degrading the training performance and sample efficiency.

3.2.2 Collective Thrust and Body Rates

When the Collective Thrust and Body Rates (CTBR) control mode is selected, the normalized outputs of the policy $\mathbf{u} \in [0, 1] \in \mathbb{R}^4$, the scaling is as follows,

$$T_{c,\text{cmd}} = u_1(T_{\max} - T_{\min}) + T_{\min} \quad (3.13)$$

$$\omega_{i,\text{cmd}} = (2u_{i+1} - 1)\omega_{i,\max}, \quad i \in \{1, 2, 3\} \quad (3.14)$$

where $\omega_{i,\max}$ indicates the maximum body rate of the respective axis, (roll, pitch, yaw). Once the rates are calculated, they can be converted to torque values,

$$\boldsymbol{\tau}_{\text{cmd}} = k_p \frac{(\boldsymbol{\omega}_{\text{cmd}} - \boldsymbol{\omega})}{dt} J \quad (3.15)$$

additionally, a proportional gain k_p can be introduced to adjust the stiffness of the conversion. Given the commanded collective thrust and torques, the motor thrusts can be obtained from Equation 3.6 and then the same procedure of Section 3.2.1 is applied during the forward and backward passes.

3.2.3 Linear Velocities and Heading Rate

The highest level of abstraction that has been implemented consists on commanding Linear Velocities and Heading Rate (LVHR). With this control mode, the policy outputs are scaled as,

$$v_{i,\text{cmd}} = (2u_i - 1)v_{i,\max}, \quad i \in \{1, 2, 3\} \quad (3.16)$$

$$\omega_{3,\text{cmd}} = \dot{\eta}_{\text{cmd}} = (2u_4 - 1)\dot{\eta}_{\max} \quad (3.17)$$

where η denotes the heading angle and $v_{i,\text{cmd}}$ represents the commanded linear velocity along each body axis. Mapping these control commands to thrust commands is less direct than in the previous methods. Instead, the collective thrust and torques are obtained by applying a transformation in the Special Orthogonal group $\text{SO}(3)$. The transformations proceed in the following manner.

First, the acceleration vector is computed as,

$$\mathbf{a}_{\text{cmd}} = k_v(\mathbf{v}_{\text{cmd}} - \mathbf{v}) + g\hat{\mathbf{e}}_3 \quad (3.18)$$

where k_v is a proportional gain on the velocity. Note that the acceleration along the third body axis is compensated for gravity; $\hat{\mathbf{e}}_3$ denotes the unit vector along this axis. Then, the force vector is the UAV's mass times the commanded acceleration vector, and the commanded collective thrust vector is the projection of the force vector onto the $\hat{\mathbf{b}}_3$ axis.

$$\mathbf{F} = m\mathbf{a}_{\text{cmd}} \quad (3.19)$$

$$T_{c,\text{cmd}} = \mathbf{F} \cdot \hat{\mathbf{b}}_3 \quad (3.20)$$

To obtain the commanded body torques, the heading vector is obtained by projecting the heading angle into the world plane defined by $\text{span}\{\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2\}$. The heading vector, serves an auxiliary vector to compute $\hat{\mathbf{b}}_2$ and $\hat{\mathbf{b}}_1$.

$$\mathbf{h} = [\cos \eta \quad \sin \eta \quad 0]^\top \quad (3.21)$$

$$\hat{\mathbf{b}}_2 = \frac{\hat{\mathbf{b}}_3 \times \mathbf{h}}{\|\hat{\mathbf{b}}_3 \times \mathbf{h}\|} \quad (3.22)$$

$$\hat{\mathbf{b}}_1 = \hat{\mathbf{b}}_2 \times \hat{\mathbf{b}}_3 \quad (3.23)$$

In this way, the commanded orientation is defined by $\mathbf{R}_{\text{cmd}} = [\hat{\mathbf{b}}_1 \quad \hat{\mathbf{b}}_2 \quad \hat{\mathbf{b}}_3]$ and the error between the commanded orientation and the current orientation is computed as in [39].

$$\mathbf{e}_R = \frac{1}{2} \left(\mathbf{R}_{\text{cmd}}^\top \mathbf{R} - \mathbf{R}^\top \mathbf{R}_{\text{cmd}} \right)^\vee \quad (3.24)$$

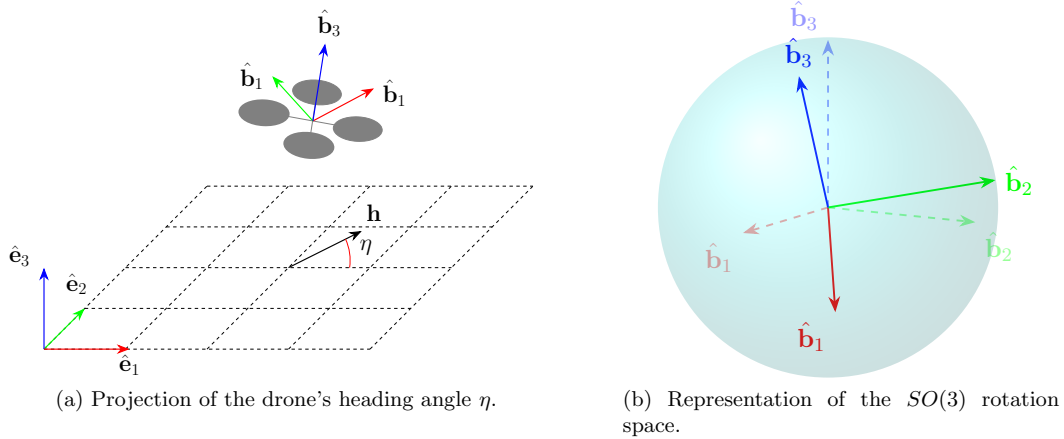


Figure 3.2: Geometric interpretation of Equations 3.12–3.15

The complete derivation of Equation 3.24 can be found in [39]. To provide intuition, note that if $\mathbf{R}_{\text{cmd}} = \mathbf{R}$, then $\mathbf{R}_{\text{cmd}}^\top \mathbf{R} = \mathbf{I}$ and the error is zero. Otherwise, the matrix $\mathbf{R}_{\text{cmd}}^\top \mathbf{R}$ represents the relative rotation between the current and desired orientations. The skew-symmetric part of the matrix $\frac{1}{2}(\mathbf{R}_{\text{cmd}}^\top \mathbf{R} - \mathbf{R}^\top \mathbf{R}_{\text{cmd}})$, encodes the axis and magnitude of the rotation error. The vee map $(\cdot)^\vee$ then converts this skew-symmetric matrix into a vector in $\mathbb{R}^3$, corresponding to the rotation error expressed in the Lie algebra $\mathfrak{so}(3)$, that is to say, $^\vee: \mathfrak{so}(3) \rightarrow \mathbb{R}^3$. Then, the factor $\frac{1}{2}$ ensures consistency with the standard definition of the vee operator and avoids double counting of the off-diagonal terms.

Additionally, an error on the angular velocity of the rotation is introduced as,

$$\mathbf{e}_\omega = \omega - \mathbf{R}^\top \mathbf{R}_{\text{cmd}} \omega_{\text{cmd}} \quad (3.25)$$

where $\boldsymbol{\omega}_{\text{cmd}} = [0 \ 0 \ \dot{\eta}_{\text{cmd}}]^\top$. Note that the error between the desired rotational velocity and the current rotational velocity line on different tangent planes to the unit sphere, $\dot{\mathbf{R}} \in \mathbf{T}_{\mathbf{R}}SO(3)$ and $\dot{\mathbf{R}}_{\text{cmd}} \in \mathbf{T}_{\mathbf{R}_{\text{cmd}}}SO(3)$. For this reason, as explained in [39], $\dot{\mathbf{R}}_{\text{cmd}}$ is transformed into a vector in the $\mathbf{T}_{\mathbf{R}}SO(3)$ space to be able to compare it to $\dot{\mathbf{R}}$ and obtain the error; Equation 3.25.

Finally, the torque is set by,

$$\boldsymbol{\tau}_{\text{cmd}} = -k_R \mathbf{e}_{\mathbf{R}} - k_\omega \mathbf{e}_\omega + \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} - \mathbf{J}(\hat{\boldsymbol{\omega}} \mathbf{R}^\top \mathbf{R}_{\text{cmd}} \boldsymbol{\omega}_{\text{cmd}} - \mathbf{R}^\top \mathbf{R}_{\text{cmd}} \dot{\boldsymbol{\omega}}_{\text{cmd}}) \quad (3.26)$$

where a gyroscopic and feed forward terms are added to the rotation and rotational speed errors. The *hat map* $\hat{\cdot}: \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is defined by the condition that $\hat{x}y = x \times y$ for all $x, y \in \mathbb{R}^3$. In this manner, the commanded collective thrust, Equation 3.20, and commanded torques, Equation 3.26, can be mapped to single rotor thrust commands.

Finally, the last control mode implemented is an augmentation of the last LVHR controller. With the last formulation, the policy outputs high-level commands in the form of desired linear velocities and heading rate. This higher level of abstraction simplifies the learning process, as the policy does not need to implicitly learn how differential rotor thrusts generate torques, since this mapping is handled by the geometric controller. However, this approach does not necessarily guarantee a better performance, as the policy can be significantly constrained by the tuning of the controller gains. For this reason, it is of interest to investigate how performance is affected when the policy is allowed to output the controller gains in addition to the velocity and heading rate commands.

Thus, when this control mode is selected, the policy outputs are, the same as Equation 3.16 and Equation 3.25, but the policy also outputs the gains controller gains $\{k_v, k_R, k_\omega\}$. Remember that the policy outputs are constrained from zero to one so the policy outputs are scaled to a maximum allowed gains,

$$k_{v,\text{cmd}} = u_5 k_{v,\text{max}} \quad (3.27)$$

$$k_{R,\text{cmd}} = u_6 k_{R,\text{max}} \quad (3.28)$$

$$k_{\omega,\text{cmd}} = u_7 k_{\omega,\text{max}}. \quad (3.29)$$

3.3 Training Task

The training objective considered for this thesis aims to provide an approximate solution to the Optimal Control Problems (OCPs), formulated in Section 2.1.3. Unlike traditional solvers that rely on Quadratic Programming (QP) or model-free RL which treats the system's dynamics as a black-box, we leverage the simulator's automatic differentiation engine to compute the analytical gradient from the cost function to the policy parameters.

Following this method, to achieve near optimum results, the training environment and policy architecture have to be carefully designed. In our simulation, we formulate them for the trajectory tracking and gate progress tasks in the next manner.

3.3.1 Trajectory Tracking

For the trajectory tracking task, the objective is to minimize a cumulative loss function over a finite horizon of length H , defined with respect to a reference state $\mathbf{x}_k^{\text{ref}}$ and the quadrotor state $\mathbf{x}_k$. The stage loss at each timestep k , denoted as $\mathcal{L}(\mathbf{x}_k^{\text{ref}}, \mathbf{x}_k)$, is formulated as a weighted sum of position, velocity, and attitude alignment errors, together with a penalization of the body rates to improve stability during early training.

$$\mathcal{L}(\mathbf{x}_k^{\text{ref}}, \mathbf{x}_k) = \lambda_p \|\mathbf{p}_k^{\text{ref}} - \mathbf{p}_k\|_2 + \lambda_v \|\mathbf{v}_k^{\text{ref}} - \mathbf{v}_k\|_2 + \lambda_a \left(1 - \hat{\mathbf{b}}_{3,k} \cdot \mathbf{z}_k\right) + \lambda_\omega \|\boldsymbol{\omega}_k\|_2^2 \quad (3.30)$$

The desired thrust direction $\mathbf{z}_k$ is defined as,

$$\mathbf{z}_k = \frac{\mathbf{a}_k^{\text{ref}} + \mathbf{g}}{\|\mathbf{a}_k^{\text{ref}} + \mathbf{g}\|_2}. \quad (3.31)$$

Since the simulator is batched, i.e., multiple environments run in parallel, the total cumulative cost over the horizon is defined as,

$$J_k = \frac{1}{B} \frac{1}{H} \sum_{i=1}^B \sum_{h=k}^{k+H-1} \mathcal{L}(\mathbf{x}_{h,i}^{\text{ref}}, \mathbf{x}_{h,i}) \quad (3.32)$$

where k denotes the initial timestep of the current horizon, H is the horizon length, and B is the number of parallel environments. At the end of each horizon, the gradient of the cumulative cost in Equation 3.32 with respect to the policy parameters is backpropagated through time for optimization, as described in Section 2.4.

To improve numerical stability and avoid exploding gradients, training episodes are terminated when predefined conditions are met. This prevents the accumulation of large errors, which can otherwise lead unstable learning dynamics. For this task, episode termination is based solely on the position tracking error. Specifically, an episode is terminated when the euclidean distance between the quadrotor position and the reference trajectory exceeds a predefined threshold d_{max} ,

$$d_{\text{max}} < \|\mathbf{p}_k^{\text{ref}} - \mathbf{p}_k\|_2. \quad (3.33)$$

This criterion prevents the system from exploring states that are excessively far from the desired trajectory, which are unlikely to provide useful learning signals and may lead to unstable gradient updates. If the termination condition is triggered during an episode, the quadrotor state is reset to the reference trajectory state and the loss over that horizon is not taken into account.

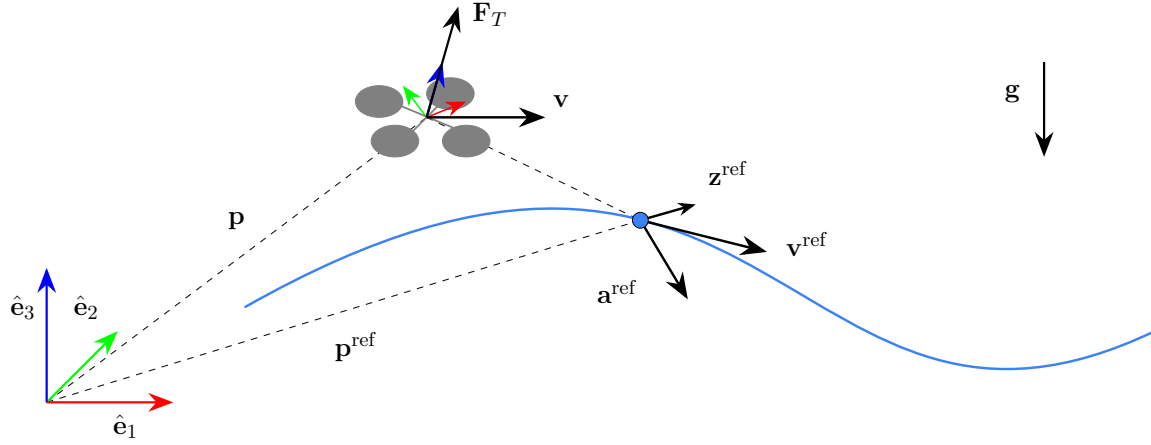


Figure 3.3: Trajectory tracking task.

Regardless of the training task, the policy network is provided with the full state of the quadrotor. For the trajectory tracking task, the observation is augmented with reference trajectory information. Specifically, instead of providing absolute position and velocity, the observation includes tracking errors together with feedforward terms from the reference trajectory. This improves learning efficiency by explicitly encoding the control objective. The observation vector is defined as:

$$\mathbf{o}_k = [\mathbf{p}_k^{\text{ref}} - \mathbf{p}_k \quad \mathbf{v}_k^{\text{ref}} - \mathbf{v}_k \quad \mathbf{a}_k^{\text{ref}} \quad \mathbf{q}_k \quad \boldsymbol{\omega}_k]^\top \in \mathbb{R}^{16} \quad (3.34)$$

The policies are first trained to track an arbitrary trajectory created by summing random harmonics as described in Equation 3.35, where $n = 3$, $\phi = 2.0$ and ω is initially set to one, but a curriculum is added in such a manner that the speed of the trajectory is incremented by a certain δ when the Root Mean Square Error (RMSE) is lower than a given threshold.

$$\mathbf{p}_k^{\text{ref}} = \sum_{i=1}^n A_i \sin(\omega_i k + \phi_i) \quad (3.35)$$

Then, the reference velocity and acceleration are approximated as the discrete derivatives of the position, where $T_s = 0.01$.

$$\mathbf{v}_k^{\text{ref}} \approx \frac{\mathbf{p}_k^{\text{ref}} - \mathbf{p}_{k-1}^{\text{ref}}}{T_s} \quad (3.36)$$

$$\mathbf{a}_k^{\text{ref}} \approx \frac{\mathbf{v}_k^{\text{ref}} - \mathbf{v}_{k-1}^{\text{ref}}}{T_s} \quad (3.37)$$

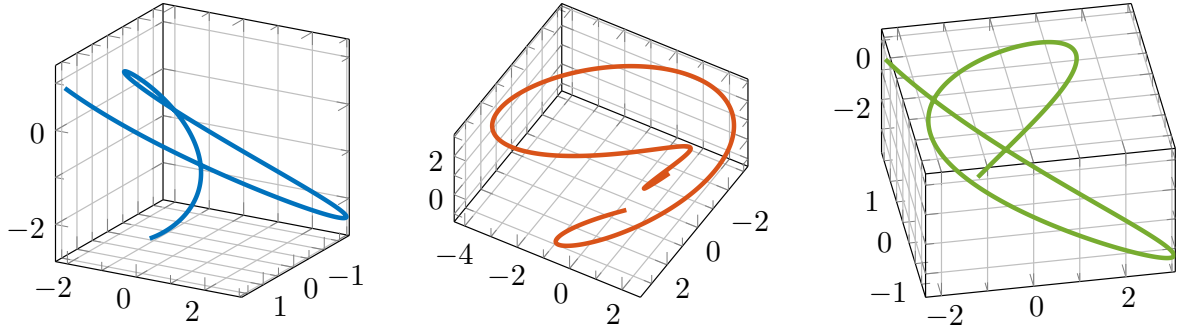


Figure 3.4: Training trajectories generated from the sum of harmonics.

3.3.2 Gate Progress

Similarly, the gate progress loss function, Equation 3.38, is also accumulated over a horizon of length H and over B parallel environments as in Equation 3.30. In this case, the loss, is function of the current target gate, the current drone's position and the last drone's position. Additionally, we penalize high body rates to encourage stability during early training.

$$\mathcal{L} = \|\mathbf{p}_k^{\mathcal{G}}\|_2 - \|\mathbf{p}_{k-1}^{\mathcal{G}}\|_2 + \lambda_{\omega} \|\boldsymbol{\omega}\|_2. \quad (3.38)$$

Note that the gate progress is expressed in the gate frame. We define the gate plane in such a manner that the $\hat{\mathbf{g}}_2$ axis is the normal and the drones are always "behind" the gate, that is to say, in the negative $\hat{\mathbf{g}}_2$ axis. Given the gate position and orientation in the world frame, the uav's relative position with respect to the gate is computed as,

$$\mathbf{p}_k^{\mathcal{G}} = \mathbf{R}_{\mathcal{G}}^{\top} (\mathbf{p}_k^{\mathcal{W}} - \mathbf{p}_{k, \text{gate}}^{\mathcal{W}}). \quad (3.39)$$

When the $\hat{\mathbf{g}}_2$ component of $\mathbf{p}_{k-1}^{\mathcal{G}}$ is negative and the $\hat{\mathbf{g}}_2$ component of $\mathbf{p}_k^{\mathcal{G}}$ is nonnegative, the gate plane is considered to be crossed.

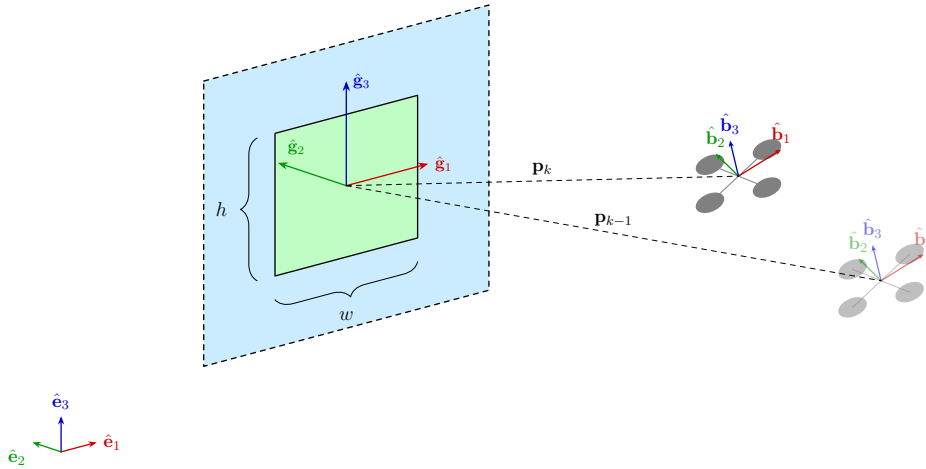


Figure 3.5: Gate traversal geometry.

During the crossing if the condition, Equation 3.40 is met, the gate is considered successfully crossed and the gate frame is changed to the next gate of the track. Otherwise, a gate crash is considered, the environment is terminated and the drone is initialized either at the starting position or behind a random gate of the track. Moreover, we set some boundaries in the world frame, if the drone crosses those boundaries we also terminate the environment to save training time and avoid exploding gradients. When the environments are terminated, their loss is not accumulated.

$$\left| p_{g_1}^{\mathcal{G}} \right| < \frac{w}{2} \quad \wedge \quad \left| p_{g_3}^{\mathcal{G}} \right| \leq \frac{h}{2}, \quad (3.40)$$

3.4 Domain Randomization

To improve the transferability of the policy and to prevent it from overfitting to a specific environment, DR is implemented in our methodology as it is a well-established strategy in the field. A domain can be defined as a specific subset of all possible Partially Observable Markov Decision Process (POMDP) environments. While a complete POMDP is formally characterized by an 8-tuple [40], we focus on a reduced 6-tuple representation that captures the essential elements for our domain definition. A standard domain is then characterized by $\mathcal{D} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, R)$ where:

- $\mathcal{S}$: denotes a set of states called the state space.
- $\mathcal{A}$: is the set of actions called the action space.
- $\mathcal{O}$: is the set of observations called the observation space.
- $\mathbf{P}(\mathbf{s}'|\mathbf{s}, \mathbf{a})$: is the probability that a given action $a \in \mathcal{A}$, at state $s \in \mathcal{S}$ will lead to a new state $s' \in \mathcal{S}$.
- $\mathbf{O}(\mathbf{o}|\mathbf{s}', \mathbf{a})$: is the observation probability, representing the probability of observing $o \in \mathcal{O}$ after taking action a and transitioning to state s' .
- $\mathbf{R}(\mathbf{s}', \mathbf{s}, \mathbf{a})$: is the immediate reward, or expected immediate reward, received for transitioning from s to s' after performing action a .

From the source domain, $\mathcal{D}_{\xi_s}$, (i.e. simulation), we can control a set of N parameters that define a configuration vector, $\boldsymbol{\xi} = [\xi_i, \xi_{i+1}, \dots, \xi_N]$. These parameters will affect the transition probability $\mathbf{P}(\mathbf{s}'|\mathbf{s}, \mathbf{a})$ of the domain. Examples of ξ_i can include:

Physical properties	$\xi_{\text{mass}}, \xi_{\text{inertia}}, \xi_{\text{drag}}, \xi_{\text{gravity}}$
Perception properties	$\xi_{\text{sensor_noise}}, \xi_{\text{sensor_latency}}$
Visual properties	$\xi_{\text{lighting}}, \xi_{\text{texture}}, \xi_{\text{fov}}$
Scene properties	$\xi_{\text{gate_pos}}, \xi_{\text{gate_size}}, \xi_{\text{gate_num}}$

When DR is applied, each configuration $\boldsymbol{\xi}$ is sampled from a bounded randomization space Ξ with a probability distribution. That is to say, $\boldsymbol{\xi} \in \Xi \subset \mathbb{R}^N$, $\boldsymbol{\xi} \sim p(\boldsymbol{\xi})$ where $\xi_i \in [\xi_i^{\text{low}}, \xi_i^{\text{high}}]$. At the start of every training episode, a new environment instance is created by sampling a new configuration vector $\boldsymbol{\xi}$ from the distribution $p(\boldsymbol{\xi})$. Therefore, the policy will interact with a new POMDP environment, i.e. a new domain $\mathcal{D}_{\boldsymbol{\xi}}$. This way, data from a variety of environments that otherwise would remain unexplored can be collected.

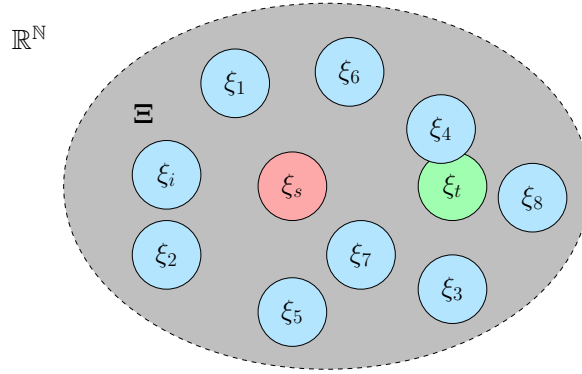


Figure 3.6: In red, the source domain. In light blue, randomized instances of the source configuration. In green, the target domain configuration.

During training, the neural network weights and biases θ are optimized to minimize the expected loss across a distribution of randomized domains [41]. With the aim of finding the optimal, θ^* that will lead to the optimal policy π_{θ}^* . As illustrated in Figure 3.6, the policy is exposed to a variety of domains to promote generalization and improve robustness, increasing the likelihood of successful transfer to the real environment. However, this strategy further reduces the sample efficiency of RL and increases training time. Therefore, it is essential to properly bound the randomization space Ξ , so that it encompasses only the minimum necessary range to include the target domain.

In order to efficiently bound this randomization space while still allowing for enough randomness a two stage randomization is implemented. Inspired by [42], first a design informed randomization is applied. This ensures that sensible variations of the nominal platform are predominant in the randomization space. Thus avoiding that unfeasible designs such as a quadrotor with unfeasible Thrust to Weight Ratio (TWR) (e.g. $TWR < 1$). The authors of [42] notice that quadrotor designs, in general, follow a similar pattern where the quadrotor's armlength l is strongly correlated to its mass m , inertia $\mathbf{J}$, drag coefficients $\mathbf{C}_D$ and other physical parameters.

Hence, they propose to sample a size factor $c \sim \mathcal{U}(0, 1)$ and scale the quadrotor's physical parameters as listed below;

$$l = c(l_{\max} - l_{\min}) + l_{\min} \quad (3.41)$$

$$m = \frac{l^3 - l_{\min}^3}{l_{\max}^3 - l_{\min}^3}(m_{\max} - m_{\min}) + m_{\min} \quad (3.42)$$

$$\mathbf{J} = \frac{l^5 - l_{\min}^5}{l_{\max}^5 - l_{\min}^5}(\mathbf{J}_{\max} - \mathbf{J}_{\min}) + \mathbf{J}_{\min} \quad (3.43)$$

$$\mathbf{C}_D = \frac{l^2 - l_{\min}^2}{l_{\max}^2 - l_{\min}^2}(\mathbf{C}_{D_{\max}} - \mathbf{C}_{D_{\min}}) + \mathbf{C}_{D_{\min}} \quad (3.44)$$

TODO :Add the rest of the params

Following the scaling process, these parameters which comprise the following configuration vector

$$\boldsymbol{\xi} = [l \quad m \quad \mathbf{J} \quad \mathbf{C}_D] \quad (3.45)$$

are subjected to additional noise. Each parameter is multiplied by a scaling factor $k \sim \mathcal{U}(1 - p, 1 + p)$, where $p = 0.2$. This specific uncertainty range was selected based on experimental results from a previous project, as it demonstrated good transferability. Nonetheless it can be considered a hyperparameter of the training process and be tuned according to the situation.

3.5 Summary

To summarize the methodology, we present Figure 3.7. At each time step, the policy takes an observation and outputs control commands. Depending on the control abstraction selected by the user, the policy will output linear velocities and heading rate, collective thrust and body rates or single rotor thrust. Then, the control commands are transformed to rotor speeds and passed through a first order transfer function to simulate the motor delay. Next, rotor speed are mapped back to thrust and then to wrench values which are used to integrate the translational and rotational dynamics. Given the next state, the loss function is evaluated and the gradient from this function to the policy parameters is backpropagated to optimize the network. Note that, during the backpropagation a surrogate model is used where the motor dynamics are skipped.

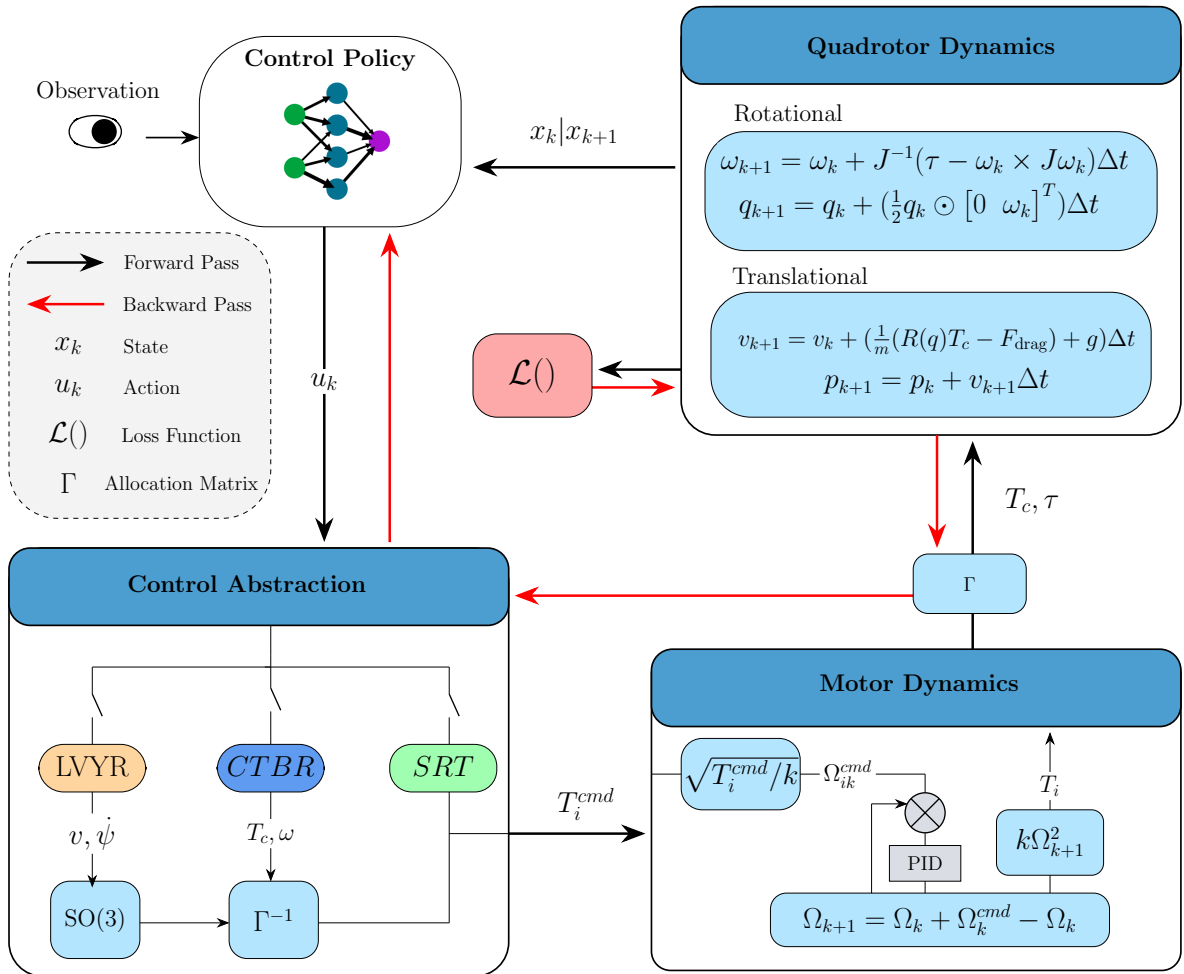


Figure 3.7: Schematics of the learning environment.

Hence, the training workflow is the following. First the user selects the training task, referred in the code as `environemnt env=[tt, pp]`, for now only the task of trajectory tracking (tt) is implemented, but other tasks such as path progress (pp) can be added. The user also selects the control abstraction/mode `cm=[srt, ctbr, lvhr, lvhr+g]` which defines the control mapping function $g_{cm}()$ from the policy outputs $\pi_{\theta}(\mathbf{o})$ to the control inputs $\mathbf{u}$. Finally, the user introduces the number of parallel environments B , the number of episodes N , the number of steps K , the optimization horizon H and other hyperparameters such as the learning rate α , randomization percentage p , termination distance threshold $d_{\max}$, etc.

Once all these parameters are set, training proceeds as follows:

Algorithm 2 Training procedure for differentiable policy optimization.

Require: Training task $\text{env} \in \{\text{tt}, \text{pp}\}$, control mode $\text{cm} \in \{\text{srt}, \text{ctbr}, \text{lvhr}, \text{lvhr}+\text{g}\}$, batch size B , episodes N , steps K , horizon H , learning rate α , simulation step $\Delta t = 0.01$ s, other hyperparams

```

1: Initialize policy  $\pi_{\theta}$ , controller  $g_{\text{cm}}$ , optimizer
2: for  $n = 1, \dots, N$  do
3:   Sample domain parameters  $\xi^{(i)} \sim p(\xi)$  for  $i = 1, \dots, B$            ▷ Domain randomization
4:   Sample reference  $\mathbf{x}_{0:K}^{\text{ref}} \leftarrow \text{getReference}(\text{env})$            ▷ Task-dependent reference
5:    $\mathbf{x}_0 \leftarrow \mathbf{x}_0^{\text{ref}}, \quad J \leftarrow 0$ 
6:   for  $k = 0, \dots, K - 1$  do
7:      $\mathbf{o}_k \leftarrow \text{getObservation}(\text{env}, \mathbf{x}_k, \mathbf{x}_k^{\text{ref}})$            ▷ Task-dependent observation
8:      $\mathbf{u}_k \leftarrow g_{\text{cm}}(\pi_{\theta}(\mathbf{o}_k))$            ▷ Policy forward pass + control mapping
9:      $\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k + \Delta t \cdot \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, \xi)$            ▷ Euler integration through dynamics
10:     $\text{term}_k \leftarrow \text{isTerminated}(\text{env}, \mathbf{x}_{k+1}, \mathbf{x}_{k+1}^{\text{ref}})$            ▷ Task-dependent termination
11:     $J += \frac{1}{B} \sum_{i=1}^B \mathcal{L}_{\text{env}}(\mathbf{x}_{k+1,i}^{\text{ref}}, \mathbf{x}_{k+1,i}) \cdot \mathbf{1}[\neg \text{term}_{k,i}]$ 
12:    Reset terminated environments:  $\mathbf{x}_{k+1,i} \leftarrow \mathbf{x}_{k+1,i}^{\text{ref}}$  for all  $i$  where  $\text{term}_{k,i}$ 
13:    if  $(k + 1) \bmod H = 0$  then           ▷ End of truncation window
14:       $\nabla_{\theta} J \leftarrow \text{backward}(J/H)$            ▷ BPTT through the computational graph
15:       $\theta \leftarrow \text{Adam.step}(\theta, \nabla_{\theta} J, \alpha)$            ▷ Parameter update
16:       $\mathbf{x}_{k+1} \leftarrow \mathbf{x}_{k+1}.\text{detach}()$            ▷ Detach state to truncate graph
17:       $J \leftarrow 0$ 
18:    end if
19:  end for
20: end for
21: return  $\theta$ 

```

Chapter 4

Results

In this chapter, the results of our methodology are presented and evaluated. The definitions of the assessment metrics used throughout the chapter can be found in Sec. A.2.

4.1 Control Abstraction Evaluation

In this section, we evaluate how the implemented control abstractions affect the sample efficiency, racing performance, as well as the transferability and robustness of the learned control policies.

4.1.1 Sample Efficiency

To evaluate sample efficiency, the four policies are trained for 40 million steps under identical conditions, including the same randomization settings, loss function, network architecture, learning rate, and hyperparameters. We then analyze how the loss evolves throughout training.

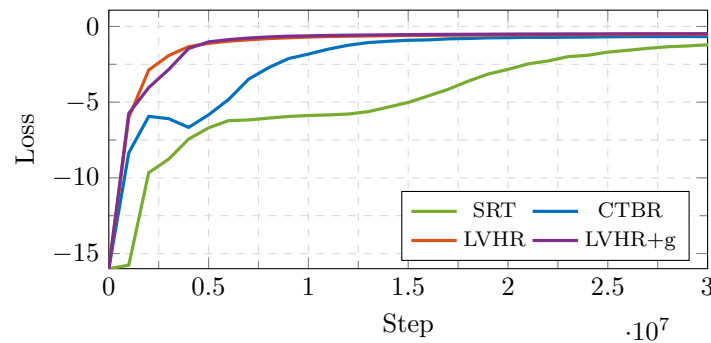


Figure 4.1: Loss versus time step for the different control abstractions.

From Figure 4.1, it can be observed that, as discussed in ??, the SRT policy requires the longest training time to converge, since the network must implicitly learn the mapping between motor commands and body rates. Although this approach is slower to train, it has

the advantage of requiring no controller gains or manual tuning. In contrast, the CTBR policy converges significantly faster, reaching convergence after approximately 1.5 million steps. The geometric control abstractions, LVHR and LVHR+g, converge even faster, requiring only around 0.5 million steps. Despite their improved convergence speed, its worth noting that latter policies rely on carefully tuned gains to ensure stable behavior, which can be time consuming and task-dependant.

4.1.2 Performance

To evaluate performance, the policies are further trained on a specific racing track. The selected track is commonly used in the literature, as it contains straight sections, high-curvature segments, and a split-S gate configuration, making it suitable for benchmarking agile flight performance. This circuit is commonly referred to as the UZH track, as it was designed by the University of Zurich.

To generate trajectories passing through the gates, the time-optimal gate traversal planner proposed in [43] is used. The planner generates trajectories that account for both the gate geometry and the quadrotor dynamics. Training directly on the optimal trajectory consistently resulted in failure to track the reference. We hypothesize that the optimization becomes trapped in local minima, preventing the policy from converging to a stable solution. To facilitate learning, a curriculum-based training strategy is adopted. Initially, the trajectory speed is scaled down to 10% of the original reference speed. Once an episode achieves an RMSE below 0.15 m, the trajectory speed is increased by an additional 10%. Using this approach, the policies are able to track the trajectory at up to 60% of the original speed, which remains significantly below the optimal performance. This limitation requires further investigation, particularly regarding the design of the loss function and the neural network architecture.

Table 4.1 and Figure 4.2 present the tracking results of the four control abstractions on the UZH track at 60% of the reference speed, where the policies achieve a maximum speed of 10 m/s.

Table 4.1: Position and Velocity RMSE of the control abstractions, mean $\pm$ std over 5 runs.

RMSE	SRT	CTBR	LVHR	LVHR+g
Position [m]	0.225 ± 0.031	0.334 ± 0.008	0.132 ± 0.003	0.201 ± 0.003
Velocity [m/s]	0.190 ± 0.002	0.180 ± 0.002	0.219 ± 0.027	0.224 ± 0.008

The results presented in Figure 4.2 and Table 4.1 indicate that the LVHR control abstraction achieves the lowest position tracking error, while CTBR obtains the lowest velocity tracking error. At this suboptimal speed, all policies achieve a Success Rate (SR) of 100%, with no gate collisions occurring during the experiments.

Furthermore, the policies exhibit broadly similar behavior. The collective thrust commands follow comparable patterns across all control abstractions. The SRT and LVHR+g policies appear to lag slightly behind the reference trajectory, whereas the CTBR policy tends to move ahead of the reference and produces the most oscillatory control commands. Although the LVHR policy achieves the best position tracking performance overall, its velocity tracking error increases significantly toward the end of the trajectory.

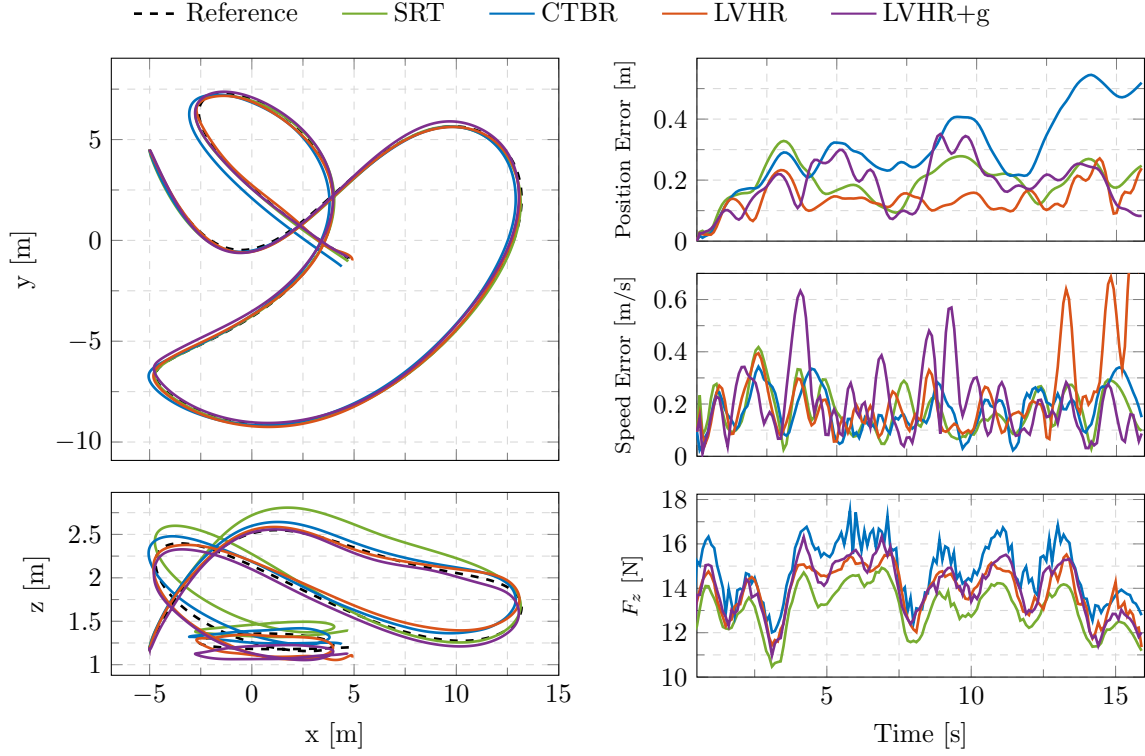


Figure 4.2: Tracking error of the different control abstractions for the UZH racing track.

Analyzing the commands of each control abstraction individually reveals information in policy behavior. For the SRT policy, in `fig:srt-action` it can be seen that the rotor commands follow a trend similar to the collective thrust profile shown in Figure 4.2. The policy successfully learns to rotate the quadrotor by generating differential thrust between the rotors. Around 12 s, high-frequency oscillations can be observed in the commands; this interval corresponds to the split-S section of the track, where aggressive rotational maneuvers are required.

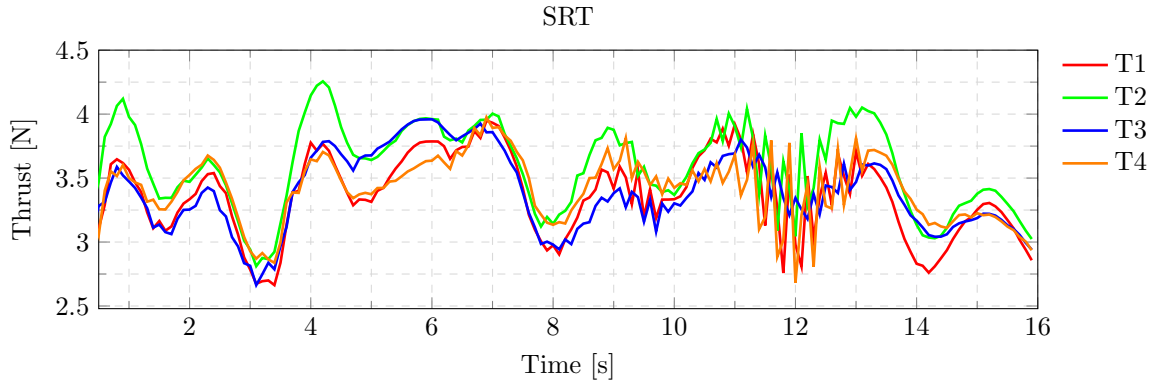


Figure 4.3: Commanded single rotor thrusts for the UZH trajectory.

From Figure 4.4, it can be observed that the control commands are relatively jittery. The pitch and roll moments are the most aggressive, reaching peak values of approximately

0.05 N · m during the split-S maneuver. In contrast, the yaw moments remain close to 0.0 N · m, since maintaining the heading aligned with the gates was not explicitly prioritized during training.

The platform has a maximum TWR of 6.9, corresponding to a maximum thrust of approximately 82 N. For the evaluated trajectory, the maximum commanded thrust is around 14 N, which corresponds to only 17% of the maximum available thrust.

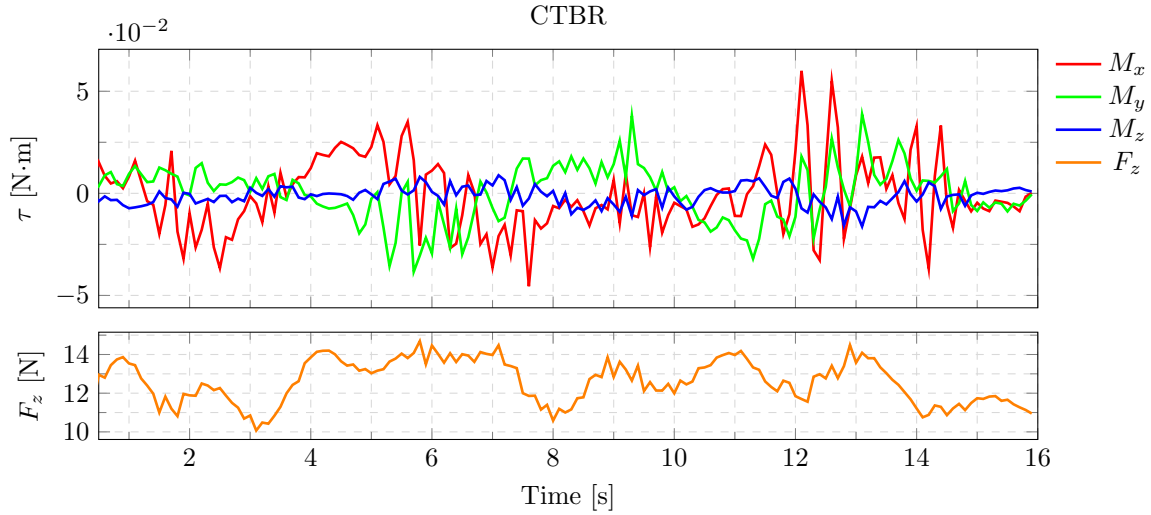


Figure 4.4: Commanded collective thrust and body rates for the UZH trajectory.

The velocity and heading-rate commands shown in Figure 4.5 exhibit a similar structure throughout the trajectory. The commanded velocities along the body x and body y axes have the largest magnitudes, while the heading rate and vertical velocity v_z remain close to zero for most of the flight.

The main advantage of this control abstraction is the smoothness of the generated commands. Additionally, as shown in Figure 4.1, this policy achieves the fastest convergence during training.

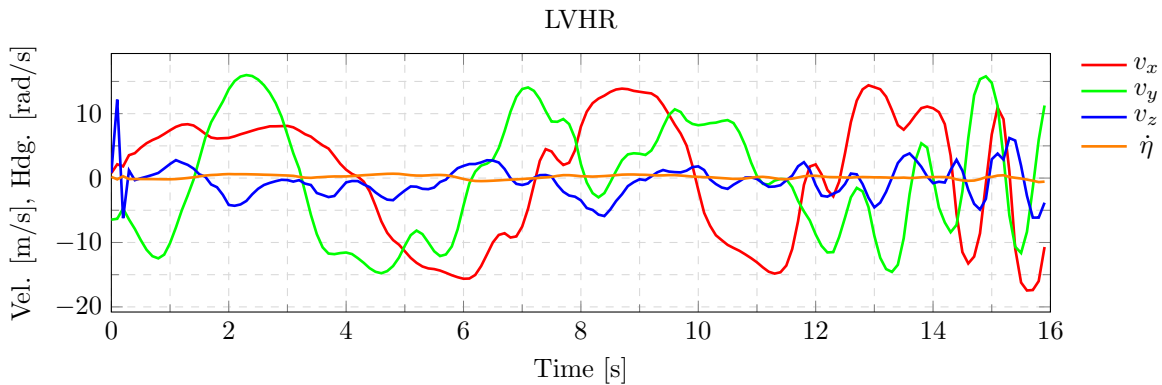


Figure 4.5: Commanded linear velocities and heading rate for the UZH trajectory.

Finally, in the current setup, augmenting the policy to output the gains of the geometric controller does not appear to provide significant improvements. The gains are initialized with

the same values used in the LVHR policy, namely $k_v = 0.6$, $k_R = 0.2$, and $k_\omega = 0.05$. However, in this case, the policy outputs scaling factors between 0 and 1 that modulate these gains.

As shown in Figure 4.6, the gains remain relatively constant throughout the trajectory, raising the question of whether the feature justified in the current setup. Nevertheless, it is difficult to draw definitive conclusions from this experiment, since the evaluated trajectory remains far from optimal and the actuators operate well below saturation for most of the flight.

Further investigation will be conducted in future work, as we believe this approach could be beneficial in more aggressive flight regimes or under varying operating conditions, potentially providing advantages over fixed-gain geometric controllers. As discussed in [44], “it is widely known that this controller design approach provides better stability and performance results for slowly varying parameters” [45].

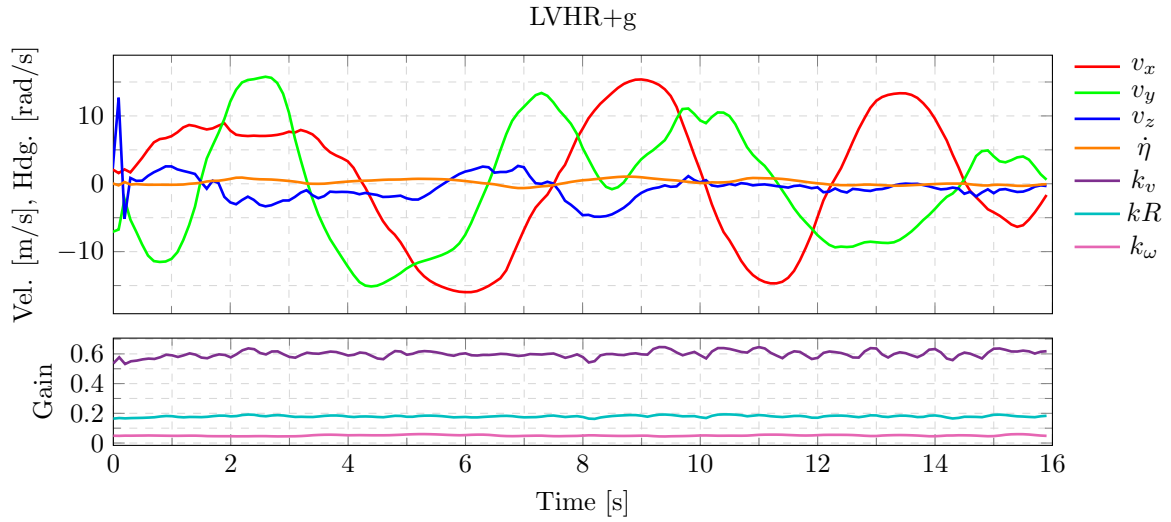


Figure 4.6: Commanded linear velocities, heading rate and controller gains for the UZH trajectory.

4.1.3 Robustness

To evaluate the robustness of the different controllers, we simulate an external wind gust. We model it as a constant acceleration of value 8 m/s^2 acting on the positive y world axis between $x \in [5, 10]$.

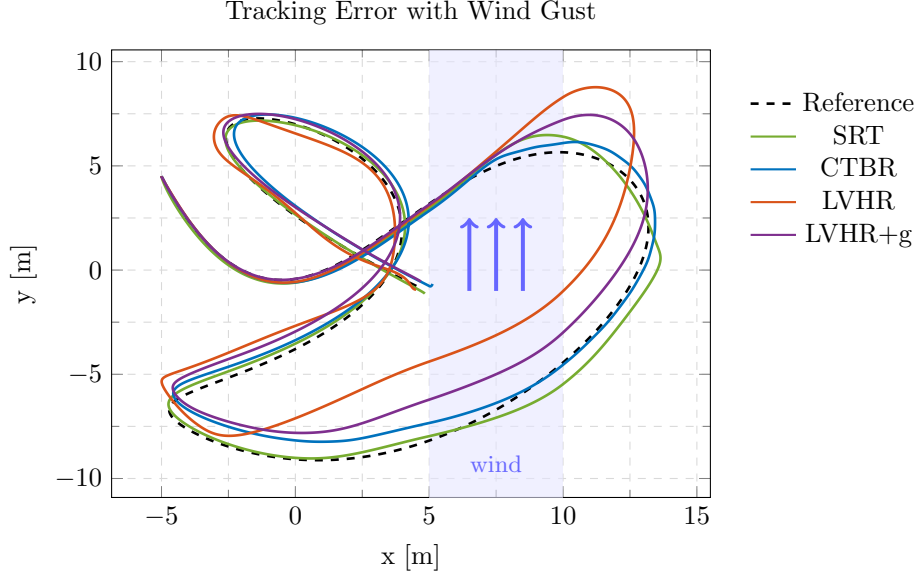


Figure 4.7: Tracking error of the different controllers with external disturbances.

Comparing Table 4.2 with Table 4.1 it can be observed that the tracking performance when considering external disturbances obviously degrades. Nonetheless, the policies remain stable and are still capable of tracking the trajectory. The control abstraction that remains the most robust is the CTBR followed by LVHR+g, SRT and finally the LVHR has the worst performance.

Table 4.2: Position and Velocity RMSE of the controllers with external disturbance, mean $\pm$ std over 5 runs.

RMSE	SRT	CTBR	LVHR	LVHR+g
Position [m]	1.229 ± 0.188	0.508 ± 0.003	1.550 ± 0.292	0.995 ± 0.036
Velocity [m/s]	1.561 ± 0.151	0.504 ± 0.002	1.568 ± 0.096	0.726 ± 0.027

4.1.4 Transferability

To evaluate how the controllers transfer to a high-fidelity simulator, we implemented a neural network controller as a ROS node in C++ and integrated into the MRS UAV system [29] within Gazebo. The CTBR policy was selected for evaluation, as this control abstraction is widely used in aerial robotics and is currently the only one supported by the MRS UAV Pixhawk 4 [46] Application Programming Interface (API). In future work, the remaining control abstractions will also be implemented within the PX4 interface, particularly the LVHR+g approach. Nevertheless, the CTBR controller provides a suitable baseline for evaluating transferability.

Using this setup, we compare the position and velocity RMSE obtained in our simulator against the results in Gazebo. In Gazebo, two configurations are considered: one using ground-truth state estimation and another using ”*passthrough*” state estimation from the MRS UAV stack which is more representative for real flight conditions.

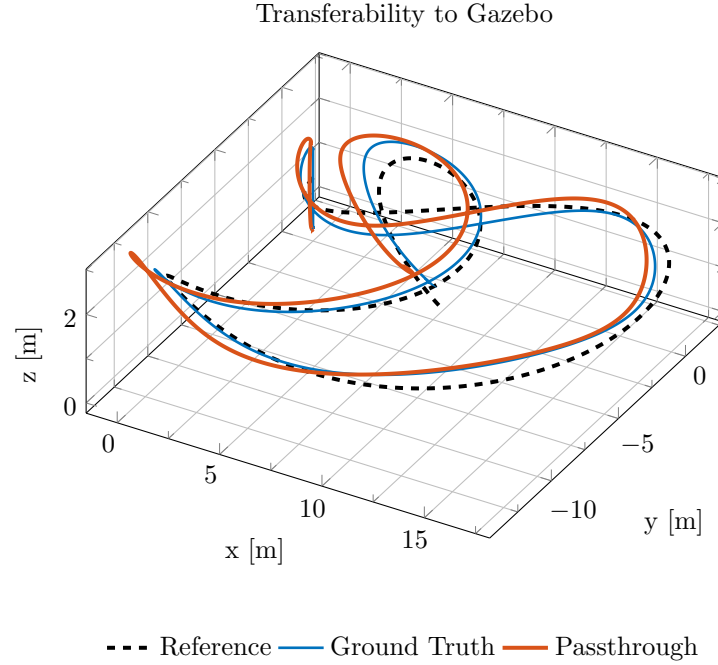


Figure 4.8: Real world flight data, UZH track trajectory tracking.

Comparing Figure 4.8 with Figure 4.2 it can be observed that there is small degradation in tracking performance when deploying the policy trained in our simulator to gazebo. Analyzing the tracking errors

Table 4.3: Position and Velocity RMSE in Gazebo.

RMSE	Ground Truth	Passthrough
Position [m]		
Velocity [m/s]		

4.2 Real World Validation

On the 16th of April 2026, during the MRS testing campaign, we conducted a real-world flight experiment. At that stage, the simulator was still on an early phase of development; actuator dynamics and aerodynamic drag were not yet modeled, and policy training relied on ground-truth state information.

As a result, there was a significant mismatch between the simulator and real flight conditions. This led to poor trajectory tracking performance, with position and velocity RMSE values of 6.11 m and 4.25 m/s, respectively. Although the overall performance was limited, Figure 4.9 shows that the general shape of the trajectory was still preserved. This suggests that

DR contributed positively to policy transferability in real-world conditions. Additionally, the experiment provided valuable insights and data that were later used to improve the simulator.

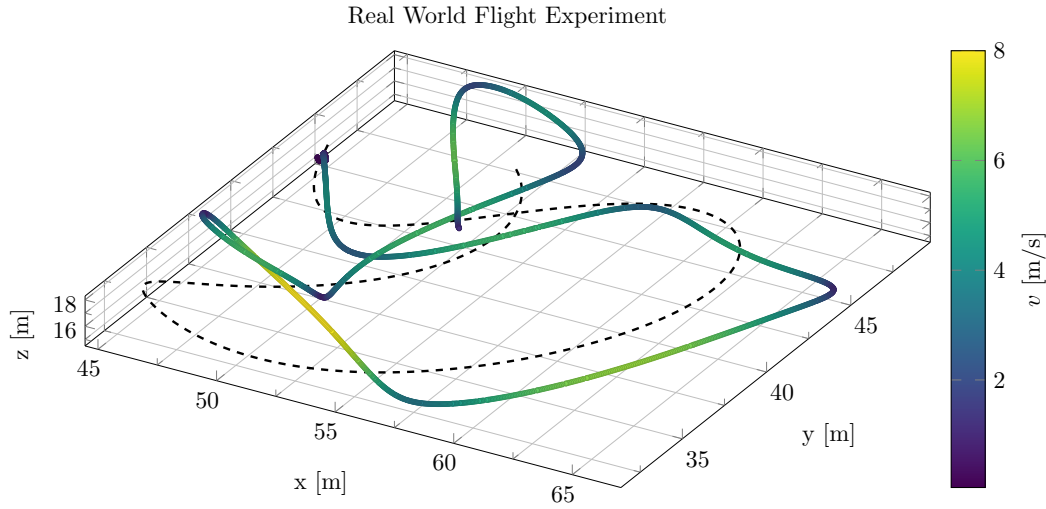


Figure 4.9: Real world flight data, UZH track trajectory tracking.

4.3 Surrogate Gradient Analysis

In this section, we analyze the effect of using surrogate gradients during the policy optimization process. The gradient is backpropagated from the loss function to the policy parameters through the chain rule. As discussed in Section 2.4, the gradients are computed numerically from the computational graph. Certain operations, such as Equation 3.11, can be highly sensitive to small perturbations, leading to unstable gradients and making the optimization process more difficult.

This behaviour can be observed in Figure 4.10, where the loss and gradient norm are plotted against the training steps for two different approaches: a simplified backpropagated model that omits motor dynamics, and the full model including the motor dynamics. At the beginning of training, both approaches exhibit similar behavior. However, the full model eventually converges to a suboptimal solution and fails to track the trajectory successfully. By analyzing the gradient norm of the full model, larger oscillations can be observed, which we believe are responsible for the unstable optimization and suboptimal convergence.

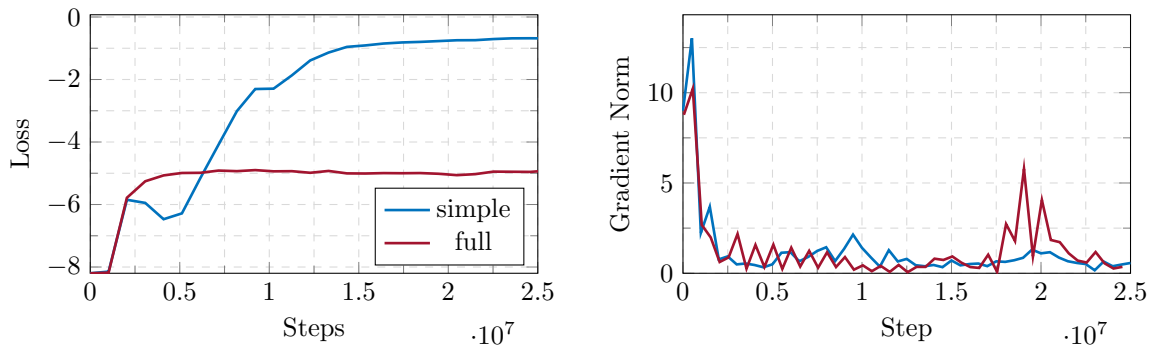


Figure 4.10: Gradients.

4.4 Model-Free and Model-Based evaluation

In this section, we compare two policies, one optimized using model-free RL with PPO, and the other using model-based RL with BPTT. The comparison is performed in terms of sample efficiency, performance, robustness and transferability.

To evaluate these methods we analyze two scenarios. Firstly, we train the policies to pass through one gate located at $\mathbf{p}_g^{\mathcal{W}} = (0.0 \ 5.0 \ 1.5)$. The UAVs' state is initialized to zero, unless for $z = 1.5$ and $q_\omega = 1.0$ which correspond to the height of the gate and identity orientation respectively. Secondly, we train for a more challenging track, an "eight" figure track. We maintain the same reward for both methods which consists in rewarding progress toward the gate and penalizing body rates as,

$$r_k = -\mathcal{L}_k = \|\mathbf{p}_{k-1}^{\mathcal{W}} - \mathbf{p}_{g_k}^{\mathcal{W}}\|_2 - \|\mathbf{p}_k^{\mathcal{W}} - \mathbf{p}_{g_k}^{\mathcal{W}}\|_2 - 0.001 \|\boldsymbol{\omega}\|_2. \quad (4.1)$$

And we run the simulation in 100 parallel environments. Note that in BPTT we aim to minimize the loss and in PPO to maximize the reward, but these are equivalent objectives.

4.4.1 Sample Efficiency

To analyze the sample efficiency we plot the mean reward over the parallel environments against the collected simulation steps for both methods.

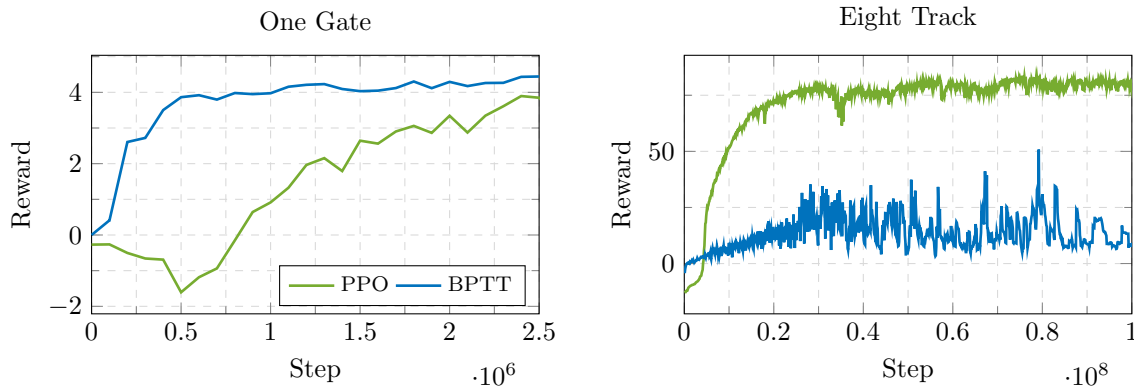


Figure 4.11: Reward against time step for PPO and BPTT

We can observe that for the simple one gate set-up, both policies converge to a reward of around 4.0. However, the BPTT policy converges in around 0.5×10^6 steps while the PPO policy around 2.5×10^6 steps. Making the BPTT method **5x** more sample efficient in this simple scenario. Nonetheless, analyzing the more challenging "eight" figure track, the PPO policy significantly outperforms the BPTT policy. The former achieves a maximum reward of 63.4 with a smooth policy updates while the latter does not surpass 50.2 and presents high variance.

We believe that this difference in the maximum achieved reward is due to the relative simplicity of our BPTT implementation compared to the actor-critic architecture and robust optimization scheme of PPO.

The BPTT policy is optimized by backpropagating gradients directly through the differentiable simulator, using a single feedforward network with no value function. While this direct approach shows higher sample efficiency for a simple scenario, when introducing multiple gates the optimal gate-traversal sequence is less smooth. This increases the risk of accumulating of bias at the truncation boundaries, causing the policy to converge prematurely to a local optimum.

In contrast, PPO leverages a critic network that maps the value of a given state, reducing the variance of the advantage estimates used to update the actor. Furthermore, the clipped surrogate objective constrains each policy update to remain within a trust region. Combined with multiple epochs of minibatch stochastic gradient updates per rollout, this yields more conservative and robust policy updates than the single-step gradient descent used in our BPTT implementation.

4.4.2 Performance

In this case, we measure performance in terms of lap time and SR over 5 runs for both policies for the "eight" figure track. To measure the best-case scenario we trained both policies with a simplified quadrotor model, not taking into account drag, motor delay, external disturbances or domain randomization and the control mode used is SRT. Note that this policies would not transfer to a real world-scenario, but this results present the highest achievable performance under ideal conditions.

Table 4.4: Performance of BPTT and PPO policies for the "eight" track (mean and std over 5 laps).

	BPTT	PPO
Lap time [s]	2.716 ± 0.086 s	1.948 ± 0.011 s
Success Rate [%]	100	100

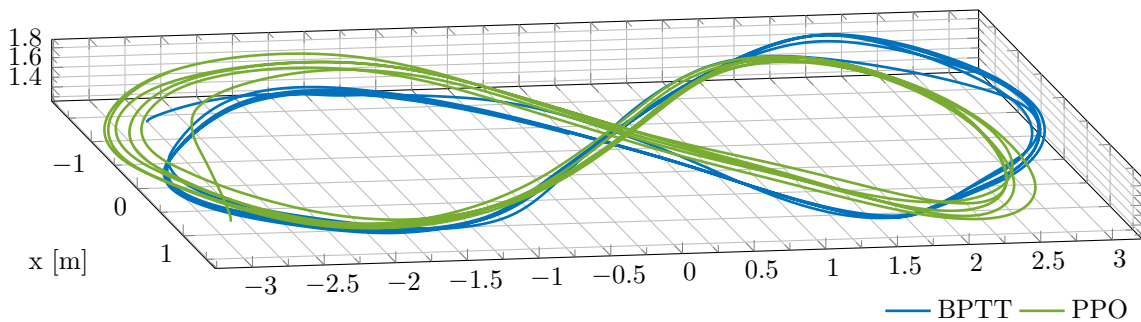


Figure 4.12: Paths traced by the policies for five laps.

In Table 4.4 we see that both policies perform the 5 laps without a single crash. And higher episode mean reward from maximizing gate progress, Figure 4.11, directly translates to faster lap times, with the PPO policy around 0.8s faster on average. This results of the PPO policy closely match the ones from [25]. Then, Figure 4.12 shows that the BPTT policy takes sharper turns than the PPO policy, which can indicate a greedier optimization approach.

4.4.3 Robustness

Next, to assess robustness we expose the environment to external disturbance, following a similar set up as in Section 4.1.3, we simulate a wind gust, which we model as an acceleration in the $-x$ axis between $y \in [-1, 1]$, see Figure 4.13. We expose the policies to different magnitudes of external acceleration and we find that the BPTT can handle up to 4 m/s^2 while the PPO policy can support up to 10 m/s^2 before dropping the perfect SR, i.e. before the first crashes.

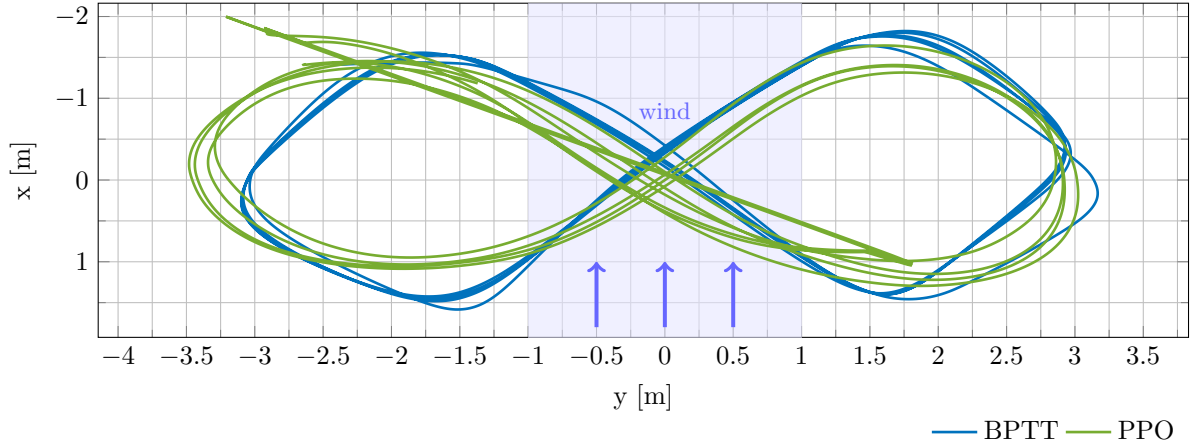


Figure 4.13: Paths traced by the policies for five laps with external disturbances.

4.4.4 Transferability

Finally, to measure transferability we perform the same experiment as in Section 4.4.2 but we train the policies with domain randomization, a model that takes into account drag and motor delay and we deploy the policies in Gazebo using the MRS stack with CTBR control mode.

4.5 Fixed Gains against Adaptive Gains

Lastly, we investigate in further detail whether having adaptive gains for the geometric controller leads to an increase in performance. The policies were optimized with PPO since it showed superior performance than our approach with BPTT and we train for the same "eight" figure track as before. Both controllers can output linear velocities in a range of $\mathbf{v} \in [-\mathbf{v}_{\max}, \mathbf{v}_{\max}]$, where $\mathbf{v}_{\max} = (10, 10, 3) \text{ m/s}$. The fixed gain policy has a proportional gain in velocity of $k_v = (3, 3, 4)$, and a gain in attitude of $k_R = (8, 8, 1)$. While the adaptive policy selects these gains dynamically at each time step in the range of

$$\begin{aligned} k_v^{\min} &= (0.5, 0.5, 0.5), & k_v^{\max} &= (6.0, 6.0, 8.0), \\ k_R^{\min} &= (1, 1, 0.2), & k_R^{\max} &= (16, 16, 2). \end{aligned}$$

Note that, contrary to Section 3.2.3, we do not include a derivative term since the policy since it experimentally showed to destabilize the policy.

4.5.1 Sample Efficiency

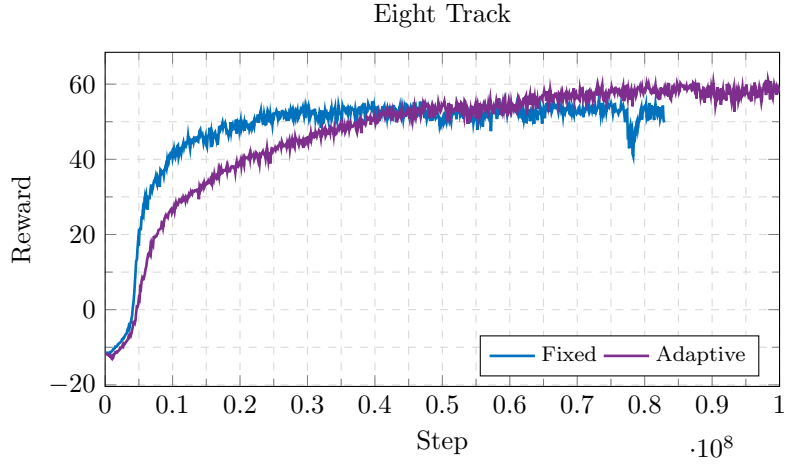


Figure 4.14: Reward against time step for the fixed and adaptive gains controllers.

4.5.2 Performance

Table 4.5: Performance of the fixed and adaptive geometric policies for the "eighth" track.

	Lap time [s]	Gates SR [%]	Track SR [%]
Fixed	2.852 ± 0.044	100	100
Adaptive	2.492 ± 0.064	100	100

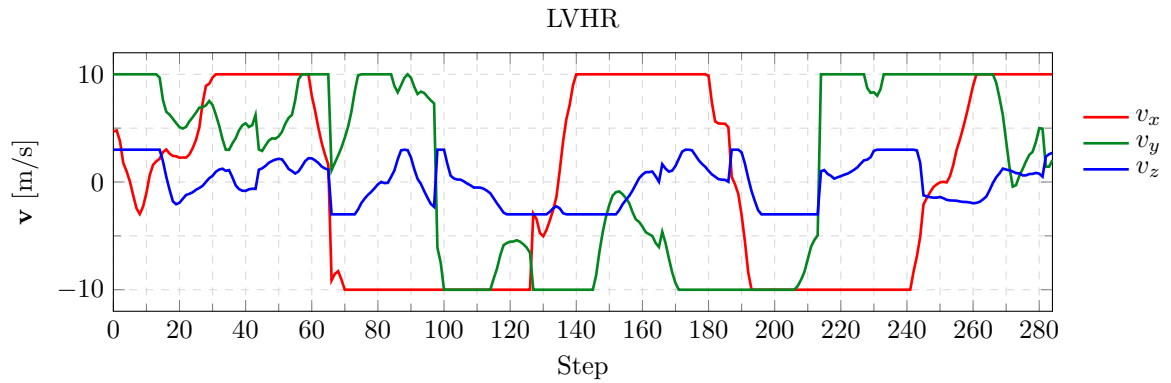


Figure 4.15: Commanded linear velocities, heading rate and gains for the eight figure track.

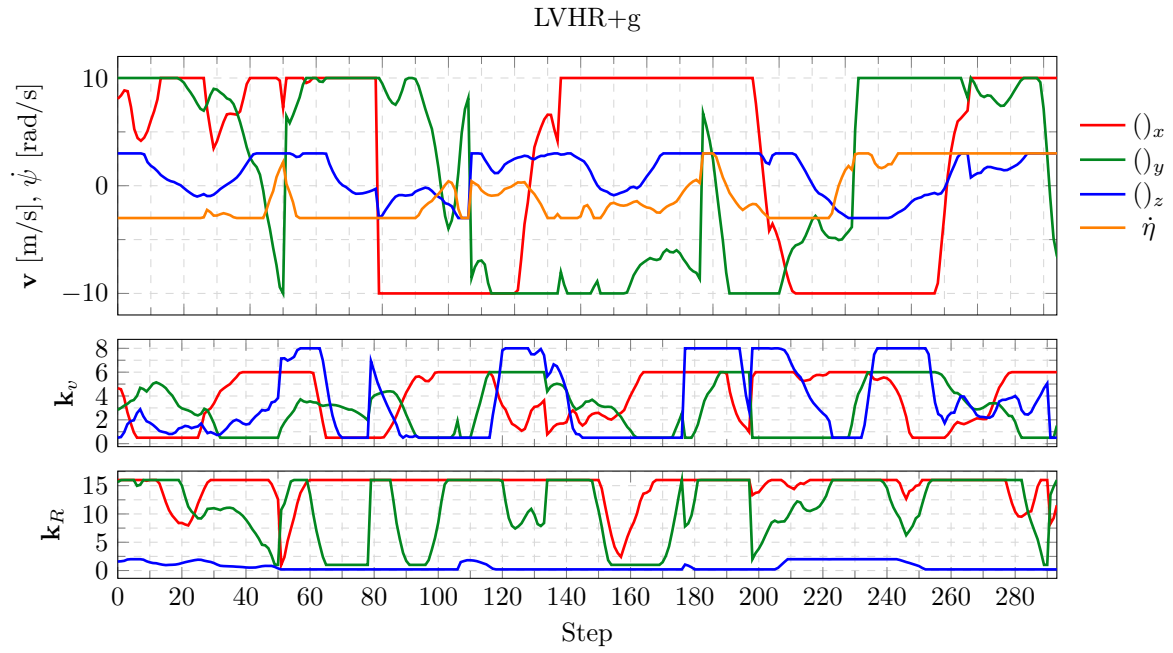


Figure 4.16: Commanded linear velocities, heading rate and gains for the eight figure track.

4.5.3 Robustness

4.5.4 Transferability

Table 4.6: Performance of the fixed and adaptive geometric policies for the "eighth" track.

	Lap time [s]	Gates SR [%]	Track SR [%]
Fixed	3.09 ± 0.12	97.5	83.33
Adaptive	3.19 ± 0.37	88.9	55.55

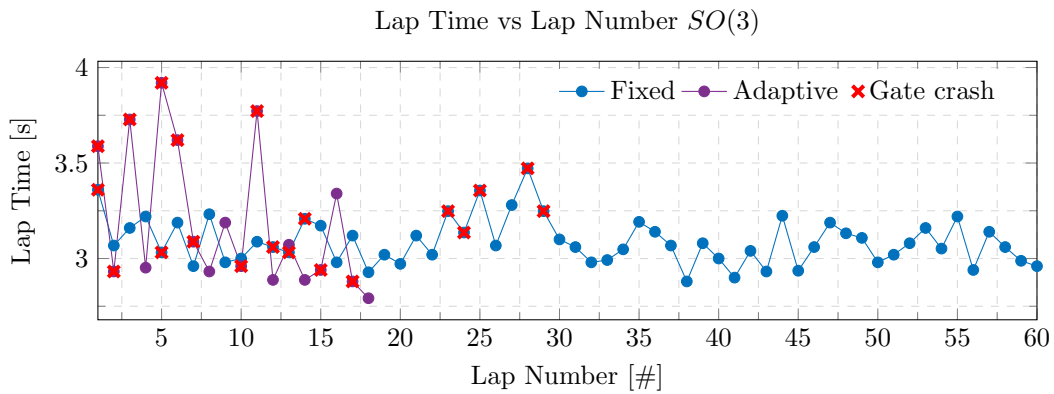


Figure 4.17: Lap times against lap number for the fixed and adaptive geometric controllers.

Chapter 5

Conclusion

In this thesis, we developed a differentiable simulator environment to efficiently train neural control policies for autonomous drone racing. The simulator leverages automatic differentiation to backpropagate the gradients from a loss function to the network parameters, increasing the sample efficiency over **7x** compared to proximal policy optimization, a standard model-free reinforcement learning algorithm. It includes domain randomization to prevent the policy from overfitting and to improve transferability to the real-world environment. It supports four control abstractions, going from direct motor commands to a geometric controller with adaptive gains. Since the motor model degrades the gradient quality, we implement a surrogate gradient for the backpropagation, in Section 4.3 we show that this improves the optimization process.

In Section 4.1.1, we find that the velocity controllers working on the special orthogonal group $SO(3)$ converge 3 times faster than the CTBR controller and 6 times faster than the SRT controller. Training is implemented in a curriculum fashion, in which we increase the speed of the trajectory when the tracking error is lower than a certain threshold. With this approach, and the loss function proposed in Section 3.3.1 we achieve a suboptimal tracking up to 10 m/s corresponding to only 60% of the optimal racing trajectory. The control abstraction that obtains the best tracking error is the LVHR, the augmented version of this controller, which outputs the gains of the inner geometric controller has similar tracking results. However, in Section 4.1.3, the LVHR controller showed to be the less robust to external disturbances, while its augmented version seemed to compensate well to external disturbances. Nonetheless, with the obtained results we conclude that CTBR controller achieves the best balance in terms of sample efficiency, performance and robustness.

Additionally, during the MRS test campaign on the 16th of April 2026, we deployed the CTBR controller on a real UAV platform. At this time, the simulator did not model motor delay, aerodynamic effects or noise in state estimation, which led to a poor tracking performance with an RMSE of 6.11 m. However, the controller was still able to stably track the trajectory, which underscores the importance of using domain randomization.

Finally in Section 4.4, we compare our model-based reinforcement learning approach using BPTT to model-free reinforcement learning using PPO.

Chapter 6

Future Directions

Chapter 7

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Appendix A

Appendix A

A.1 Simulator Validation

As detailed in ??, our simulator utilizes a simplified model of quadrotor dynamics and aerodynamics. To maintain a computationally efficient and generalizable framework, we omit secondary effects such as battery depletion on thrust, complex aerodynamic effects, and the propellers rotational inertia. While these unmodeled dynamics can cumulatively lead to a significant sim-to-real gap, we aim to address this mismatch through DR. To validate the reliability of this simplified approach, we benchmark our simulator's behavior against Gazebo, a high-fidelity physics engine widely adopted by the robotics community. Specifically, we utilize the UAV models provided by the MRS UAV System within the Gazebo environment as our ground-truth reference for comparison.

We separately assess the responses of the body rates against a unit step, a variable amplitude and a variable frequency signals.

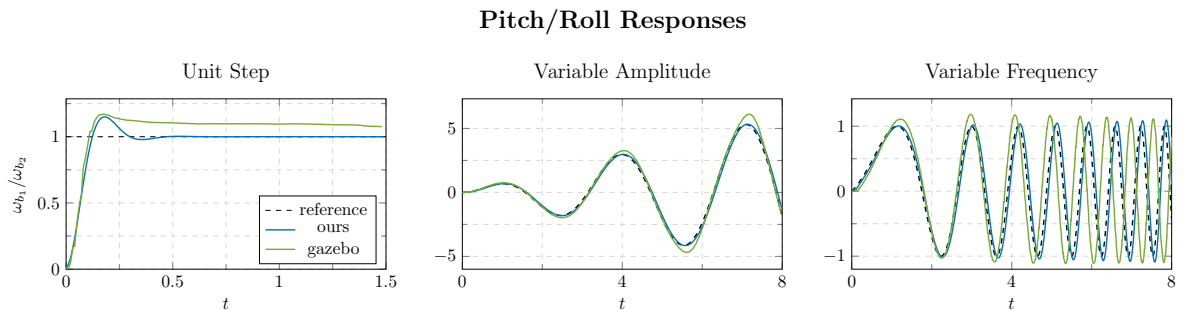


Figure A.1: Pitch/Roll responses to unit step, variable amplitude and variable frequency signals.

For the pitch and roll responses, the behavior observed in Gazebo and in our simulator is consistent. Minor discrepancies can be observed in the response of integration of the unit-step error, together with a slight overshoot in the variable-frequency signal.

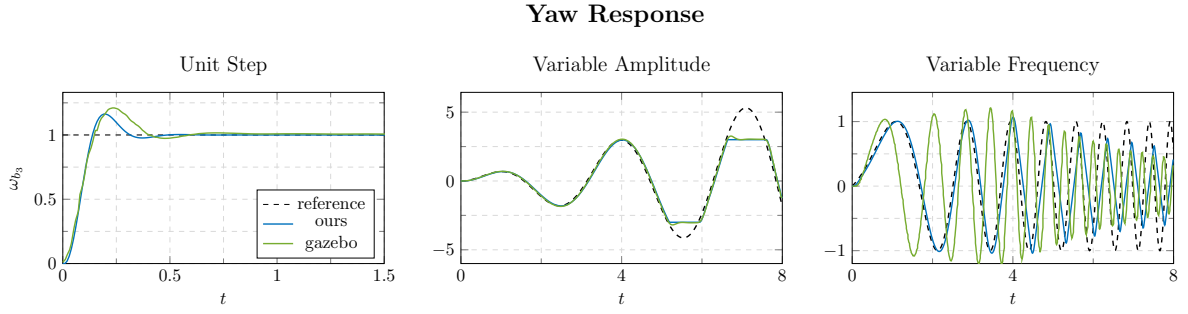


Figure A.2: Yaw response to unit step, variable amplitude and variable frequency signals.

Similarly, the yaw response are close but there is some mismatch on the variable frequency signal. In our simulator we saturate the yaw rate to ± 3 rad/s and the yaw acceleration to ± 5 rad/s² to match with Gazebo.

A.2 Evaluation Criteria

This section details the metrics and experimental framework used to evaluate the performance of the proposed differentiable physics approach against a model-free PPO. The evaluation is structured around three primary axes: training efficiency, performance in terms of lap time and success rate, and sim-to-sim transferability.

A.2.1 Sample Efficiency

One of the central claims of this thesis is that leveraging analytical gradients from differentiable simulation significantly improves sample efficiency compared to standard model-free RL algorithms. To evaluate this claim, the training convergence of the proposed differentiable physics approach is compared against PPO trained on the same task, observation space, network architecture, and control abstraction. Sample efficiency is measured as the wall-clock training time and the number of environment steps required for each method to converge to a stable policy. Convergence is defined as the point at which the cumulative loss/reward ceases to improve by more than a threshold. For a fair comparison, the randomized processes use a common seed during the training process.

A.2.2 Performance

Lap Time and Success Rate

Ultimately, the primary objective of ADR is to traverse a sequence of gates as fast as possible. The two metrics used to evaluate this objective are the lap time T_f and the SR.

The lap time T_f is defined as the time elapsed from the start of the episode until the drone passes the final gate. The success rate is defined as the ratio of successfully passed gates G_p to the total number of gates G in the racing track,

$$SR = \frac{G_p}{G} \in [0, 1]. \quad (\text{A.1})$$

A gate g is characterized by its position $\mathbf{p}_g \in \mathbb{R}^3$, orientation $\mathbf{R}_g \in SO(3)$, and dimensions (t, w, h) , corresponding to the thickness, width, and height of the gate opening, respectively. The gate frame axes are defined as,

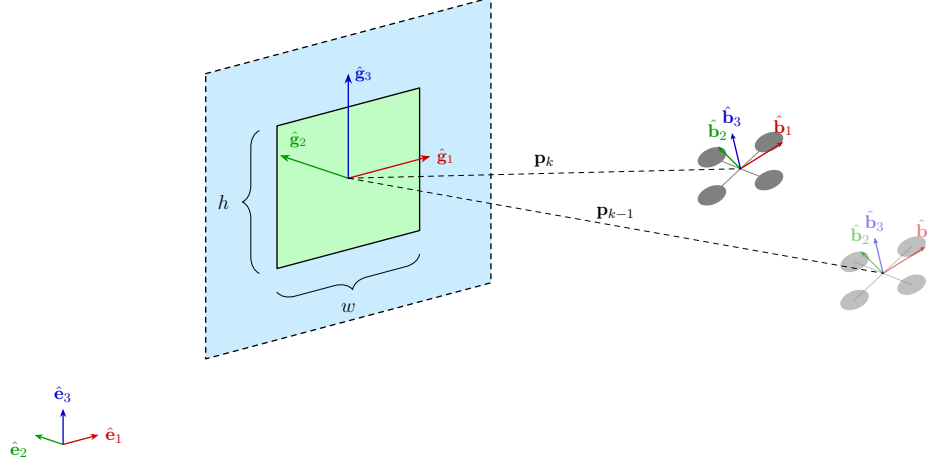


Figure A.3: Gate traversal geometry. The success region (green) is defined with respect to the gate position $\mathbf{p}_g$, orientation $\mathbf{R}_g$ and dimensions (t, w, h) . **TODO add gate dimension to the plot, t,w,h**

$$\hat{\mathbf{g}}_1 = \mathbf{R}_g \hat{\mathbf{e}}_1, \quad \hat{\mathbf{g}}_2 = \mathbf{R}_g \hat{\mathbf{e}}_2, \quad \hat{\mathbf{g}}_3 = \mathbf{R}_g \hat{\mathbf{e}}_3, \quad (\text{A.2})$$

where $\hat{\mathbf{g}}_1$, $\hat{\mathbf{g}}_2$, and $\hat{\mathbf{g}}_3$ denote the forward (normal), lateral, and vertical axes of the gate, respectively. The position of the drone relative to the gate, expressed in the gate frame, is,

$$\mathbf{p}_k^g = \mathbf{R}_g^\top (\mathbf{p}_k - \mathbf{p}_g) = [p_1^g \quad p_2^g \quad p_3^g]^\top. \quad (\text{A.3})$$

A gate g is considered successfully passed at timestep k if,

$$|p_1^g| < \frac{t}{2} \quad \wedge \quad |p_2^g| \leq \frac{w}{2} \quad \wedge \quad |p_3^g| \leq \frac{h}{2}, \quad (\text{A.4})$$

and a gate is unsuccessfully passed when,

$$|p_1^g| < \frac{t}{2} \quad \wedge \quad \left(|p_2^g| > \frac{w}{2} \vee |p_3^g| > \frac{h}{2} \right). \quad (\text{A.5})$$

Root Mean Square Error

When a reference path is used, another performance metric utilized is to compute the RMSE between the drone's position and the reference path.

A.2.3 Transferability and Robustness

Learned policies are prone to overfitting to the training environment and may fail to generalize when deployed under conditions that deviate from those seen during training, due to model mismatches and unmodeled effects. To mitigate this, DR is employed during training as described in ??.

To assess transferability, the policies trained in the proposed differentiable simulator are evaluated in a high-fidelity simulation environment using the MRS UAV stack in Gazebo [29]. Since the differentiable physics and model-free RL policies are optimized under different paradigms, the sim-to-sim performance gap is analyzed for each method, measuring the degradation in success rate SR and lap time T_f relative to the results obtained in the training simulator. Additionally, robustness to external disturbances is evaluated by exposing the trained policies to wind perturbations during deployment, without any retraining or fine-tuning.